KKT in practice: implementation repairs, measured scaling, and the graph structures that change the comparison

Evidence cutoff: 15 October 2026. This edition covers the completed implementation experiments, hierarchy and distance size ladders, current Google Benchmark campaign, and lower-density timing and untimed work studies. Their full captures passed strict reanalysis and independent numerical or counter audits. No partial campaign is used as evidence. Historical Markdown, HTML and PDF editions remain unchanged.

Repairs restored local accounting missing from the original implementation, and later experiments reduced constants. A faster KKT implementation still need not be the fastest MST solver on a given graph.

The lower-density follow-up now finds the requested **sustained strong crossover against retained-edge Borůvka** on distance-ranked attachments at m=16n: **(512,8,192]**, maintained through every measured size up to 2,097,152. Its largest-size wall/CPU ratio is **0.708792/0.708792**. The density 8 ladder reverses to mixed graph results at 4M; shuffled twins lose throughout. Kruskal beats all largest endpoints, so this is a specific useful KKT family, not an all-solver or universal asymptotic victory.

Private-eighth KKT leads retained-edge Borůvka on ordered attachments: narrowly at **1,024 vertices**, strongly at every measured size from **4,096 through 4,194,304**. Wall ratios are **0.67130 at 1M and 0.84144 at 4M**; at 4M only nine of twelve trials win. Because the smallest point already leads, no loss-to-win crossover is observed. Shuffled twins lose throughout, and no case makes KKT fastest of all six solvers by the declared medians.

Distance-ranked attachments provide a second favorable synthetic arrangement: **0.68067 wall and 0.68042 CPU** against retained B at 1M, with all 72 pairs winning both clocks. Again, an initial narrow 1K lead prevents an observed retained-B crossover. Topology-identical shuffled twins lose every pair: weight organization matters, and spatial/random losses remain.

The earlier half-policy ratios at 65,536, 1,048,576 and 4,194,304 were **1.1181, 0.9674 and 1.0786** against retained B: an intermediate lead reversed. Against original B they were **1.6442, 1.0111 and 0.8861**, a strong bracket for that comparator. These distinct policies/campaigns cannot be spliced into one curve.

Google Benchmark, rather than GoogleTest, supplies the requested complexity estimator. Historical family exponents of 1.113-1.185 and the current executable's universally selected N log N labels provide no linear-runtime confirmation. Neither finite result refutes the expected linear-operation bound.

These are implementation- and workload-specific measurements, not universal crossover predictions or an independently published verdict on an encyclopedic “galactic algorithm” classification.

1. Which implementation, input size and problem?

All comparisons solve the minimum spanning tree problem: find a connected acyclic set of edges with minimum total weight. Disconnected correctness fixtures use the corresponding minimum spanning forest objective. Dijkstra was explicitly replaced by Borůvka in the requested comparison. Dijkstra minimizes distances from a source, which is a different objective: with AB=2, BC=3 and AC=4, the shortest-path tree from A uses AB and AC, while the MST uses AB and BC.

Let **n** be the represented vertex count, **m** the number of undirected edge records, and **N=n+m** the input size. On a fixed-density family with m=dn, linear in N is also linear in n. On a complete graph, m=n(n−1)/2, so linear input work is quadratic in vertices. The distinction also matters after contraction: parallel records can remain even though the number of quotient vertices becomes small.

“Current KKT” needs a build qualification. The experiments introduce opt-in paths, and not every measured optimization is combined with every other one:

Path discussed	Quotient contraction	Verifier	Private sampling	Evidence at this cutoff
Repaired ordinary behavior	Deduplicate parallel pairs	Tuple input, individual requests	Scalar half	Historical repaired/default measurements
Retention experiment	Retain crossing records	Tuple, individual requests	Scalar half	Paired screen and stronger-control follow-up complete
Direct-Edge experiment	Retain crossing records	Direct Edge input, individual requests	Scalar half	Paired experiment and density screen complete
Packed sampling	Retain crossing records	Direct Edge, individual requests	Packed half	Separate paired experiment complete
Compressed queries	Retain crossing records	Direct Edge, shared requests	Scalar half	Paired experiment, common-family and larger screens complete
Half versus eighth	Retain crossing records	Direct Edge, shared requests	Scalar half versus scalar eighth	Paired timing, work diagnostics and seven-size extension complete
Packed plus compressed	Retain crossing records	Direct Edge, shared requests	Packed half	Source preparation only

The ordinary build still selects deduplication, tuple input, individual queries and scalar half sampling. Public legacy sampling is separately preserved. Having experimental source in a checkout does not mean that path is selected, promoted, or represented by historical default timings. Fresh reference builds also matter: a source refactor can change generated code even when it preserves behavior.

To reproduce the current opt-in path, the reviewed build recipe uses `KKT_EXPERIMENT_RETAIN_PARALLEL` and `KKT_EXPERIMENT_EDGE_VIEW`, adds `KKT_EXPERIMENT_COMPRESSED_QUERIES` when compiling `verifier/verifier_v2.cpp`, and adds `KKT_EXPERIMENT_EIGHTH_SAMPLING` when compiling the eighth variant's `kkt/kkt.cpp`. The executable at `bin/eighth-sampling/timing-build/eighth/retention_control` is an ignored local build artifact, not part of the publication bundle. Its tracked build record and freeze retain its hash identity and recipe evidence. Use the study's bounded reproduction procedure and full provenance checks; these flags alone do not reproduce its environment or measurements. This remains an experimental build, not a promotion into the ordinary default.

All local report links now target evidence in the current eighth repository, including the committed copy of the completed packed study and its archived tested helper. The copy receipt records that transfer; its original-machine native paths and build dependencies remain unchanged. Final tracked/copied-link completeness still requires its separate gate, so local path placement alone is not a portable native rebuild claim. The historical report and each study's own freeze remain the authority for their recorded sources and timings.

2. Why the original program paid more than the algorithm required

The recognizable KKT structure was already present in original revision 693c6a5: contract, sample, recursively solve the sample, filter heavy edges, and recursively solve what remains. The problem was that some ostensibly local operations depended on the original problem size or performed superlinear preprocessing.

Original edge IDs controlled recursive storage

Let R=1+max(original edge ID). Contractions initialized R-sized tracking arrays, recursive calls allocated R-sized result storage, and merging scanned the same range. Even when IDs were dense, R described the root edge list rather than the current recursive input. A child containing a dozen edges could still initialize and scan storage associated with millions of original edges.

This invalidates the premise that a recursive node's own work is bounded by a constant times its own input. A weighted hierarchical path can retain a main branch through logarithmically many contraction levels. Because a tree has no removable heavy edges, repeatedly scanning an original range proportional to n contributes superlinear bookkeeping. Sparse IDs make the defect more severe: changing an ID can enlarge R without adding any represented graph data. Negative IDs and overflowing the old integer maximum-ID calculation also expose correctness boundaries.

Revision `60c65b4` made original IDs payload. Selected IDs accumulate in vectors, and local tracking uses positions in the current edge array. Legacy maximum-ID metadata was widened and stopped governing allocation. The essential repair was the domain of bookkeeping, not a blanket preference for one container type. A bitmap is cheap only when the range being cleared and scanned is appropriate to the problem being solved.

The median lookup table was expensive to build

The heavy-edge verifier reduces a tree to a full branching tree and uses a table over depth masks. The old table builder repeatedly enumerated subset members. In an h-bit universe, the total number of included bits across all subsets is $h \cdot 2^{(h-1)}$. One slice of the old preprocessing therefore already required $\Omega(h \cdot 2^h)$ work. For an equal-depth full binary instance with $L=2^h$ leaves, that is $\Omega(L \log L)$.

Revision `312954e` replaced repeated enumeration with a recurrence that removes extreme set bits and reuses an already computed inner-mask answer. Each mask takes constant word work. The table is linear in the relevant reduced input because this reduction gives every original leaf the same depth and every internal node at least two children: $L \geq 2^h$, so the allocated mask table has at most $2L$ entries.

Equal leaf depth is a necessary part of that argument. An arbitrary binary tree can be highly unbalanced. Merely saying “full branching tree” without checking the property produced by the source would leave a gap in the accounting. The repair and structural regression tests are linked from the historical report; this defect can be identified from source without fitting timings.

Sampling restarted the same global stream

The old sampler called `srand(seed)` again inside recursive sampling. It replayed prefixes and changed process-global state. Revision `ccb833f` introduced one continuing private MT19937 stream per invocation. Repeating a whole invocation with a chosen seed remains intentional, but recursion no longer restarts the same prefix at every sample.

This change restores the intended discipline of taking fresh choices conditional on the recursive problem. It does not make a finite deterministic PRNG a literal sequence of independent mathematical coins. Nor was repeated seeding, by itself, evidence that the returned MST was wrong: the sampling analysis and the correctness of a particular returned tree are separate questions.

Benchmark repairs were a different category

The old benchmark fixed n=10,000 while increasing m from 1.25 million to 20 million. That primarily swept density. It also reused a default generation seed, lacked independent certification of measured outputs, and excluded KKT's tuple-to-internal input conversion from its loop. Those limitations restrict what its curves establish; they should not be misreported as algorithmic speedups.

Other repairs addressed a Prim/KKT global Edge-type collision, distinct test seeds, exact unique-MST ID checks and reproducible builds. Comparing only an integer sum was insufficient because the legacy cost helper can overflow. Exact IDs and tuple metadata now establish a much stronger result than an accidentally matching sum.

After the structural fixes, six accepted constant reductions removed local heavy-edge hashing, linked query lists, selected-edge adjacency construction, sampled-forest recovery hashing, temporary graph copies and repeated final hash-table bucket growth. Their commits are `4eb3ca1`, `2749ced`, `2e37b4e`, `af12309`, `ba39977` and `2d4fbf1`. Each preserved a stated invariant such as caller IDs, edge order, sampling draws or returned membership. A minimum-edge cost cache was rejected: doubling that per-vertex working set brought some small gains but about a 6% largest-case regression.

The historical repair study measured, for example, **62.66 ms → 13.99 ms** at n=4,096 and m=524,288, a paired 4.54× improvement. The separate practical campaign improved all 39 case medians, with final/baseline ratios about 0.393-0.969. Those are useful results, but their inputs and timing boundaries differ. They cannot be multiplied with later gains to create an unmeasured end-to-end speedup. Complete original repairs, accepted changes and rejected results remain in the historical evidence.

3. Expected linear work is a conditional source claim

KKT combines safe-edge contraction with sampled-forest filtering. If an edge is heavier than the maximum edge on a forest path connecting its endpoints, the cycle property excludes it from the MST. The sampled forest need not itself be part of the final MST. When endpoints lie in different forest components, that edge remains light and cannot be removed by this test.

The primary paper supplies the crucial sampling bound: if each edge is selected independently with probability p and F is the sample's MSF, the expected number of F-light edges is at most n/p. Its stated contraction deletes loops, isolates and duplicate quotient edges, and its algorithm chooses p=1/2. Those are the published ingredients; retaining all crossing parallel records is a separate implementation extension. Karger, Klein and Tarjan, §§2-4.

The local source argument uses the implementation contract examined here: valid loop-free undirected inputs, distinct edge IDs and weights, and represented integer bounds; normalization handles isolated vertices. These are conditions of this implementation study, not universal restrictions on MST theory: equal weights can be ordered consistently. After two contractions, let $q \le n/4$ be the surviving vertex count and $r \le m$ the crossing records. Sampling each record independently gives expected sample size pr. Processing distinct-weight records in increasing order shows why the light-edge bound also works with parallels and a disconnected sample: a light record is recognized before its own coin is exposed, and each accepted light record adds one forest edge. The expected accepted count is at most q. Thus the expected light count is at most q/p. Finished components and isolates cannot enlarge the surviving quotient.

Assume all nonrecursive work is at most $c(m+n)$, the verifier is linear, and the next random choices obey the stated conditional model. Seeking $E[T(n,m)] \le Am+Bn$ gives sufficient conditions

$$A \ge c / (1 - p),\ B \ge 2c + A / (2p).$$

For half sampling, A=2c and B=4c suffice; for eighth sampling, A=8c/7 and B=46c/7 suffice. These are our accounting bounds from the fixed-probability note, not measured constants. Every fixed p strictly between zero and one admits such a bound. Arbitrarily changing p with size requires another

argument. Empty dispatches are charged to their parents, with an O(1) addition for an entirely empty input.

This explains both why the original root-sized bookkeeping was unacceptable and why fewer sampled edges are not automatically better. Reducing p lowers the sample branch's expected edge count but weakens the later light-edge bound. Both branches must be solved, and the parent still pays for verification. The simple proxy pr+q/p captures that tradeoff but omits actual branch costs, forest shape, allocation and data access.

The verifier also needs scrutiny. The audit covers geometric reduction, mask-table construction, LCA preprocessing and aggregate path-query work. It finds no obvious remaining asymptotic factor in those audited phases, but it is not a formal proof of every C++ execution. Public result hashing assumes expected constant-cost operations; masks, counts and memory must remain representable. The mathematical model does not supply a machine-independent runtime coefficient or a new high-probability theorem for this retained-parallel implementation.

The most defensible theoretical statement is therefore **expected O(n+m) under the documented input, primitive-cost and randomness assumptions**. It does not mean all finite time/N curves must be flat, all families must favor KKT, or every comparator must actually realize its worst-case upper bound.

4. What Google Benchmark actually measured

GoogleTest supplies correctness and structural checks. Google Benchmark supplies the requested complexity fit through `SetComplexityN(n+m)` and `Complexity(oAuto)`. The repository pins version 1.5.2. Its implementation selects a model using CPU measurements and then fits real/manual measurements to that selected model. Automatic choices come from a small zero-intercept menu: constant, log N, N, N log N, N^2 and N^3. The printed normalized RMS describes residual fit quality; a selected label is not a complexity proof. Pinned Google Benchmark implementation.

Historical diagnostics also fitted manual times to cN^α and restricted windows. The five-family run used infrastructure c9ed4ca, unchanged production `2d4fbf1` and algorithm/boundary checks against `1a3fc3c`: nine doubling sizes through 262,144, three graphs, two algorithms and three batches of three calls, totaling 810 records/2,430 calls. Its Bazel O3/NDEBUG profile differs from the later native profile.

Historical family	Full-window α	Upper-window α	Counted work/N, first → last	Built-in KKT label
Bipartite small side 16	1.126	1.173	7.31 → 7.20	N log N
Ring of 16-cliques	1.185	1.309	9.19 → 9.15	N^2
3D grid, distance ranks	1.113	1.100	17.66 → 18.50	N log N
90% attached leaves	1.115	1.102	11.43 → 14.05	N log N
Degree-eight regular	1.157	1.167	24.68 → 32.45	N log N

None received N: the clique's free-power exponent 1.185 accompanies an N^2 menu choice. Earlier exponents were about 1.001 for a star, 1.175 for sparse random and 1.379 for denser random. These finite fits neither refute an expected bound nor establish persistent superlinearity.

Historical normalized-time figure: the earlier repaired source, not the current experimental executable. It remains in the supplementary gallery below.

Almost-flat small-cover/clique work per N accompanies rising time/N; these counters do not explain timing or measure cache/allocator costs. Other families increase work per input: small dense quotients deduplicate far more than larger fixed-degree quotients.

Historical fixed registration order and ratio statistics differ from the later rotating studies. Nontrivial startup load averages are not CPU-utilization percentages. Windows, intercepts, memory effects and host history limit all fitted labels; no observations are removed.

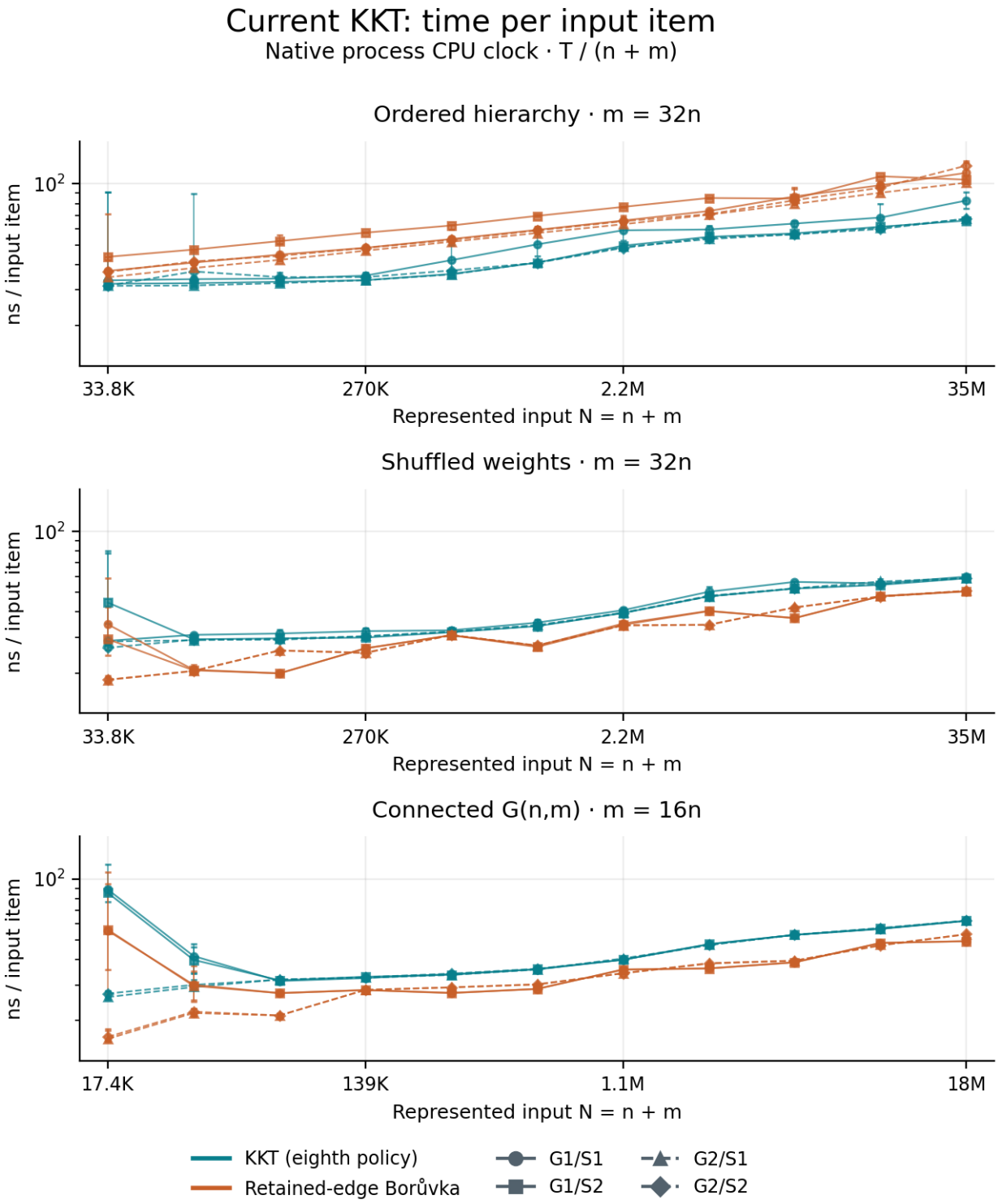
Current experimental executable: all choices are N log N

The completed current study uses the same eighth-sampling production objects as the comparison studies, with a new pinned Google Benchmark main. It contains three families, two algorithms and eleven doubling sizes n=1,024…1,048,576; N=n+m, with m=32n for ordered/shuffled attachments and m=16n for connected G(n,m). All 2,376 calls (1,584 timed, 792 warmups) and 792 certificates remain. Two graph seeds and two sampler labels create four KKT curves per family; retained-edge Borůvka ignores the sampler label and repeats only two distinct graph inputs. Repetitions also reuse inputs. G1 sizes ascend and G2 descend, so graph identity and size order are confounded.

Google selects N log N for all 24 curves, including all twelve KKT curves. The independent six-model analysis likewise favors N log N in all 192 curve/clock/window combinations. The table retains each exported choice; RMS is the library-CPU residual percentage, not uncertainty.

Family / algorithm	G1/S1	G1/S2	G2/S1	G2/S2	CPU RMS range
Ordered hierarchy / KKT	N log N	N log N	N log N	N log N	3.62-16.91%
Ordered hierarchy / Retained B	N log N	N log N	N log N	N log N	8.11-24.11%
Shuffled twin / KKT	N log N	N log N	N log N	N log N	3.22-3.77%
Shuffled twin / Retained B	N log N	N log N	N log N	N log N	6.43-8.55%
Connected G(n,m) / KKT	N log N	N log N	N log N	N log N	5.25-5.61%
Connected G(n,m) / Retained B	N log N	N log N	N log N	N log N	7.12-10.67%

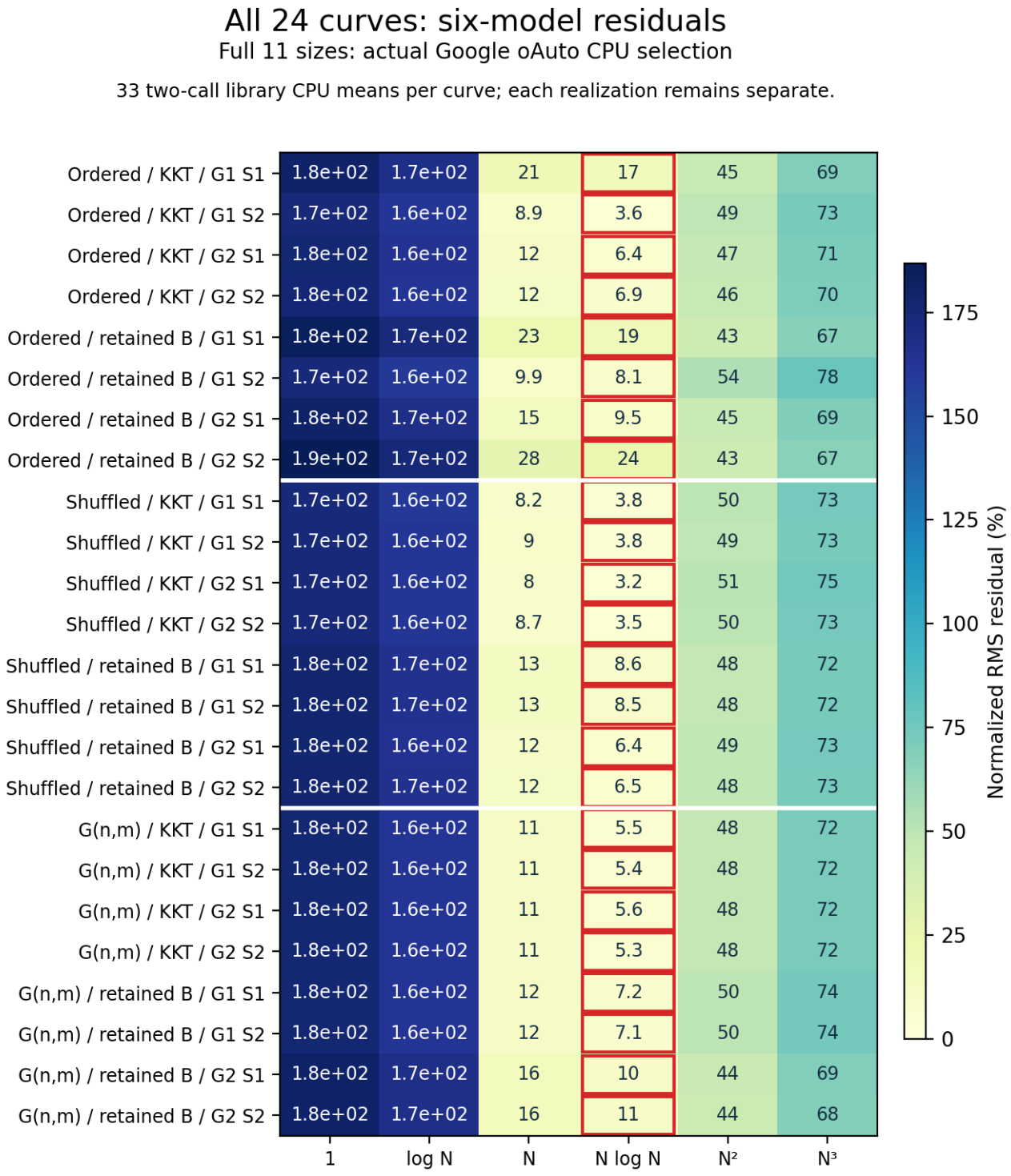
Normalized native CPU time rises at the large end for both algorithms in every family. Ordered KKT is favorable against retained B over much of this range; shuffled and random KKT remain more expensive at the large endpoints. Small points are irregular: random KKT at n=1,024 has CPU/N of 85.56-88.90 ns for G1 versus 26.11-27.17 ns for G2, with elevated small G1 points also in the control. No point is trimmed and graph/order confounding prevents causal attribution. Full ranges are observed calls, not confidence intervals. Separately scheduled absolute medians do not establish a paired crossover.



Current native process-CPU time per represented input; all four labels on two graphs

All four clocks remain separate: native steady elapsed time, native process CPU, Google manual time and Google CPU. Google's CPU interval includes different loop/timer work from the native interval above. Its CPU-selected model determines both CPU and manual coefficients. Only CPU RMS is

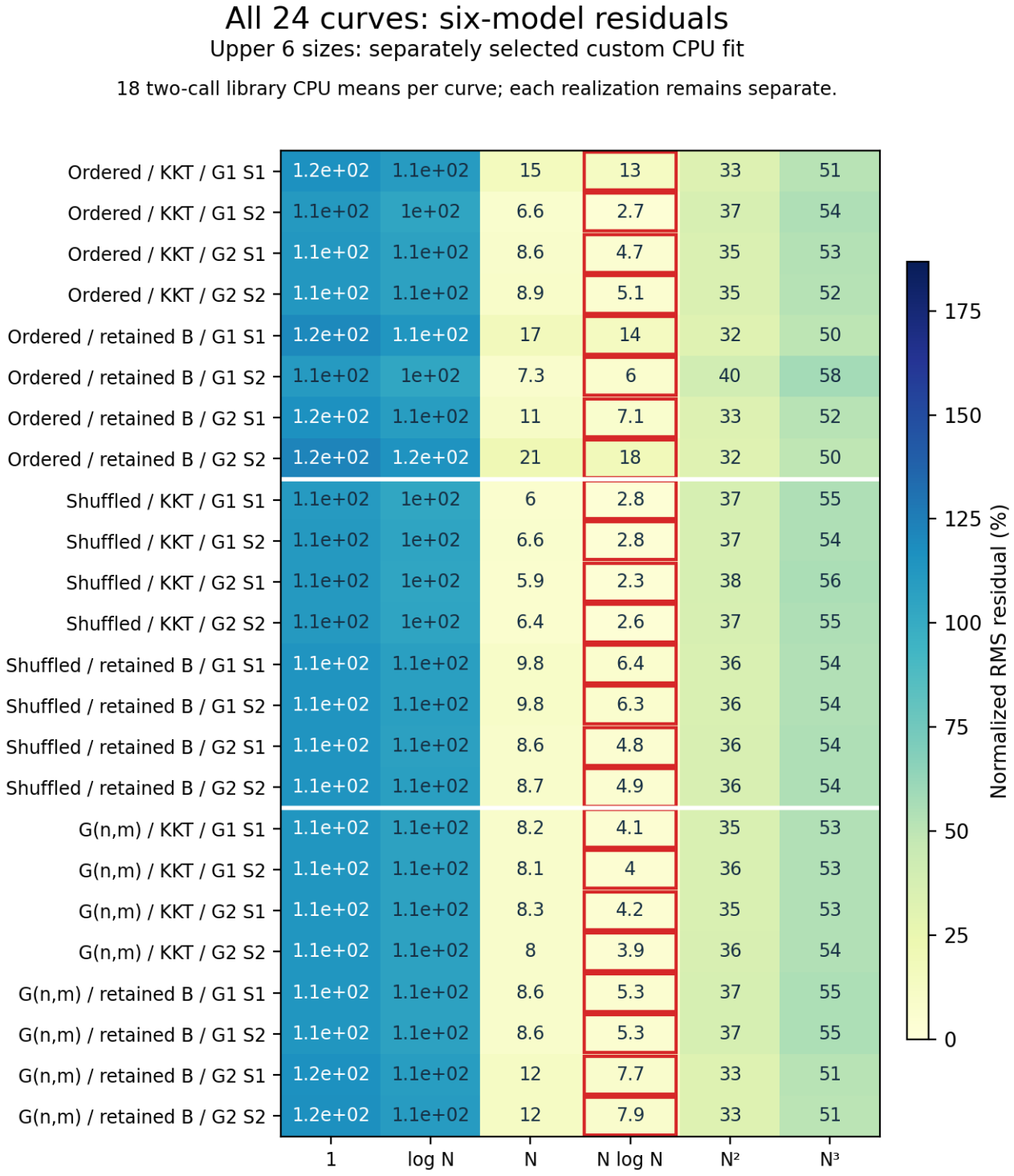
exported; manual RMS is independently reconstructed, and each coefficient's unit is validated. The full-window page retains all six alternatives for every curve.



All 24 full-window choices and residuals, including every competing model

Full11 uses 33 two-call means per curve/clock. The predetermined upper6 uses 18 means over n=32,768…1,048,576; its highlighted minimum is a custom fit, not another Google result. All

upper-window choices also remain N log N. KKT's descriptive native-CPU log-log exponents over that window are 1.1267-1.1313 ordered, 1.1470-1.1591 shuffled and 1.1599-1.1616 random. Those slopes are hash-bound analysis outputs, not independently re-audited quantities.



Red outline: custom upper-window minimum, not a built-in result.

Google shares its CPU-selected model with the manual coefficient; CPU RMS only.

Both pages share color limits. Exact four-clock/all-model values remain in CSV.

Residual labels are rounded for display. Fit labels do not prove asymptotic complexity.

The predetermined upper-six-size model comparison; a custom sensitivity analysis

The independent audit reconstructs all 3,168 means, 1,152 fits and 24 Google selections/coefficient semantics; size summaries, slopes, flags and ordinary library aggregates have a narrower hash-binding check. Zero-intercept models, large-duration points and only two graphs limit identification. Fixed m/n does not determine a general two-variable bound. These results do not establish linear measured runtime, nor do they refute expected O(m+n) operations.

There are zero clock flags and logged sleep overlaps. The native campaign took 58.291004 minutes, with peak sampled owned-group RSS 4.3583 GiB and zero unavailable observations; all groups exited. Sleep inhibition was unverified. Older Google macOS metadata may report Mach tick rate as frequency, and its scaling flag does not establish stable thermals. All eleven figures passed original-resolution QA. Complete clocks, coefficients, all models and raw evidence are linked in the study.

5. A fair practical comparison needs six implementations

The six solvers are KKT, Kruskal, Prim and three Borůvkas. Original B scans the original tuple graph; dedup B rebuilds a simple quotient; retained B repeats KKT's cheaper retained-parallel step with comparable ID accumulation/result construction. This shared-kernel control tests whether sampling, recursion and verification repay their cost. A win against deduplication alone cannot answer that question; these controls are not proofs of optimal engineering.

The Prim source keeps at most one tentative `std::set` entry per vertex and scans 2m adjacency records. Only successful improvements update the set: O(m+(n+u) log n), u≤m. In the hierarchy with its forced cheap path, visited core vertices form an interval; chords are checked without necessarily improving candidates. We did not measure u or its CPU cost. Prim's worst-case O(m log n) label therefore does not force a finite crossover.

Katriel, Sanders and Träff's 2003 **I-Max-Filter** already balances pm and n/p with p=√(n/m). This different, expected O(m+n log n) algorithm uses Prim order and interval maxima. Its random, vertex-distance, stretched-geometric and Prim-update-heavy experiments show filtering helping difficult dense inputs while random weights can favor Prim. Those machines/results are not ours; the square-root balance is prior art, and neither study establishes superiority over all MST implementations. Author-hosted paper, sections 1-3.

All start from the same tuple list. KKT and the two ID-returning continuations include internal conversion, temporary-input destruction and output construction; returned-result checking/destruction are outside. Tuple-returning solvers retain their APIs. Generation, certification and validation are untimed. Another application's input representation could change rankings, especially Prim's.

Each six-solver child has one checked warmup and six timed repetitions, rotating every solver through every position. Case medians use paired ratios, never ratios of separate medians. Two graphs/one sampler give twelve ratios; two graphs/two samplers give twenty-four. Repetitions reuse graphs and samples, not independent draws.

Wall/CPU clocks and full ranges are retained, not treated as independent machines or confidence intervals. Reversed schedules counterbalance order but leave seed/order/host-history confounding. Unchanged-control drift is reported without subtracting it from KKT or normalizing results.

The prespecified strong-crossover rule requires every graph/realization median on both clocks to exceed one earlier, then be ≤0.90 later and at every remaining measured size. Exact fractions decide thresholds. Otherwise outcomes may be narrow, mixed, losing or already leading without a losing endpoint. A bracket is not an interpolated transition; a final-point win establishes no later persistence. Reversals remain visible even if another bracket later qualifies. All comparator columns remain.

6. Four later reductions in constants, kept separate

Retain parallels when deduplication buys little

Retaining crossing records in input order preserves safe selections, numbering and loop/isolate removal while replacing buckets/linked traversal with exact reservation and at most three scans rather than four. Public simple-quotient contraction is unchanged. In the ten-case screen, all 100 KKT pairs improved on both clocks; large ordered times were about 48% of reference. One root's deduplication removed only 0.07% of nearly sixteen million crossing records. This variant also changes output order, draw-to-edge assignment and potentially recursion.

The fresh shared-step control reversed the interpretation: retained B won all 120 pairs on both clocks. At n=2,097,152, density 16, ordered KKT/retained B was 1.120 despite KKT/dedup B being 0.381. A cheaper contraction benefits simpler continuations too.

Read existing Edge records directly

The Edge overload removes the pre-verifier tuple copy. Only the KKT call-site object differs; both binaries share the refactored verifier. Gates preserve 63 counters and reduce visits/materializations by exactly the verifier-input edge count. Ten medians improve 2.9-6.8%, with 115/120 wins per clock; retained B still wins 120/120. The fresh tuple reference is not the earlier historical binary.

Consume all bits in each private random word

Packed half sampling consumes 32 bits per MT19937 word in edge order, discarding unused suffixes at call boundaries: ceil(r/32) words for a nonempty r-edge call. Public sampling and half-size reservation stay unchanged, but equal seeds yield different private samples.

The packed study improves six medians 8.1-17.7% and all 144 pairs on both clocks; control case ratios 0.9926-1.0112 do not eliminate confounding. Small ordered KKT wins 24/24 retained-B trials with pooled ratio 0.8957, but one realization exceeds 0.90, making the declared lead narrow. The larger ratio 1.0048 gives no size crossover. Its tested helper and evidence are locally archived under the copy receipt; native binaries/machine dependencies were not copied.

Recursive sample volume falls in 13 pairs, rises in 11, and has median ratio 0.999378. Aggregate engine-word *bounds*, not observed totals, are 0.02760-0.03243 of scalar counts. This shows lower engine consumption without systematic recursive-volume collapse, not 32× solver improvement or CPU attribution. Its uncompressed-verifier gains cannot be multiplied into the next experiment.

Share repeated verifier requests

Compression answers repeated lower-vertex/ancestor-depth path requests once, preserving each heavy mark. The experiment changes only the verifier object, sharing seven others including KKT. Million-vertex gains are 8.4-14.6%; all three small medians regress 2.6-7.5%. Ordered KKT/retained B changes 1.1167→0.9707, a narrow transition, not a 10% lead. Original B wins every direct pair.

Isolated experiment	KKT implementation effect	Stronger-control conclusion
Retained parallels	All 100 trial pairs improve; large ordered ratios about 0.48	Fresh retained B wins 120/120
Direct Edge input	Ten medians improve 2.9-6.8%; 115/120 trial wins	Retained B wins 120/120
Packed half sampling	Six medians improve 8.1-17.7%; 144/144 trial wins	Small hierarchy lead weakens to large parity
Compressed requests	Large gains 8.4-14.6%; every small median regresses	Narrow hierarchy transition; later size reversal

Different equivalence guarantees and fresh references preclude multiplying these percentages into one cumulative speedup.

7. Why query compression helps without solving the whole problem

In one connected verifier-tree call, Q counts proper-ancestor endpoint requests including duplicates; U counts distinct lower-vertex/depth pairs. A reduced tree with B vertices and height h gives

$$U \le \min(Q, \Sigma_v \operatorname{depth}(v)) \le \min(Q, B \cdot h).$$

Production endpoints are r equal-depth leaves: m_0 non-loop within-component query edges give $Q=2m_0$ and $U \le \min(Q,rh)$. This is per verifier call, not original-graph average degree. Loops/cross-component edges contribute no such pair; $Q=U=0$ makes U/Q undefined.

Immutable R[v] marks local answers; propagated D[v] also carries descendant requests needed by traversal. The original traversal/order computes one answer per R bit, then a graph scan uses popcount ranks and the unchanged strict heavy comparison. Merging R and D would be incorrect. The constructed sharing illustration has 24 requests/four answers. Independent positional/forest-path gates cover duplicate/negative IDs, extreme weights, loops, disconnected forests, malformed lookups and actual dispatch. Equivalent outputs and source reasoning preserve filtering; not every intermediate forest was recorded. Details remain in the complete study.

Per completed solve, cumulative U/Q is **0.2696-0.8738%**, median 0.5194%: ratios of sums, not per-call distributions or global request deduplication. Here Q,U, query edges M, connected calls T and reduced vertices BΣ are summed over that solve's connected verifier-tree calls. Exact accounting is $4Q \to 2Q+U$ query visits; initialized slots fall by $M+Q-U+T$, verifier-edge visits by M, and verifier-vertex visits rise by BΣ. The four identities hold; 61 other fields match. Median compressed/reference ratios are **0.4236 initialized slots, 0.5013 query visits, 0.6673 verifier-edge visits**, with roughly doubled verifier-vertex visits.

These heterogeneous slots are not bytes. Cached LCAs survive longer, replacing per-edge maxima during traversal; local masks, offsets, answers, recursion and capacity still matter. Duplicate computation/scatter decreases, but depth work/lookups increase and answer accesses are not globally sequential. No hardware profile establishes locality or peak-RSS savings; even regressing small cases have over 99% duplicate requests.

In the compressed half-policy study, the million-vertex hierarchy makes retained B perform 19 rounds and 17.4846-17.4849 original-edge input volumes, versus eight rounds/7.3168-7.5177 for shuffled twins. Shuffled queries share more, yet KKT/retained B is 1.62 rather than 0.97. Less competitor contraction work can outweigh greater KKT request reuse.

8. Weight arrangements that make repeated contraction expensive

The ordered attachment construction isolates a particular mechanism. Write n=4c: each core vertex has three very cheap leaf edges, and the c core vertices have a forced path plus distinct nonpath chords. In `leaf_hierarchy4`, all path edges are cheaper than all chords; path weight levels force dyadic sibling-block merges. `leaf_distance4` instead orders every core edge by an ancestry tier, so some shortcuts can join the MST. The shuffled twin permutes core ranks while preserving topology, IDs, labels, physical order and cheap leaves.

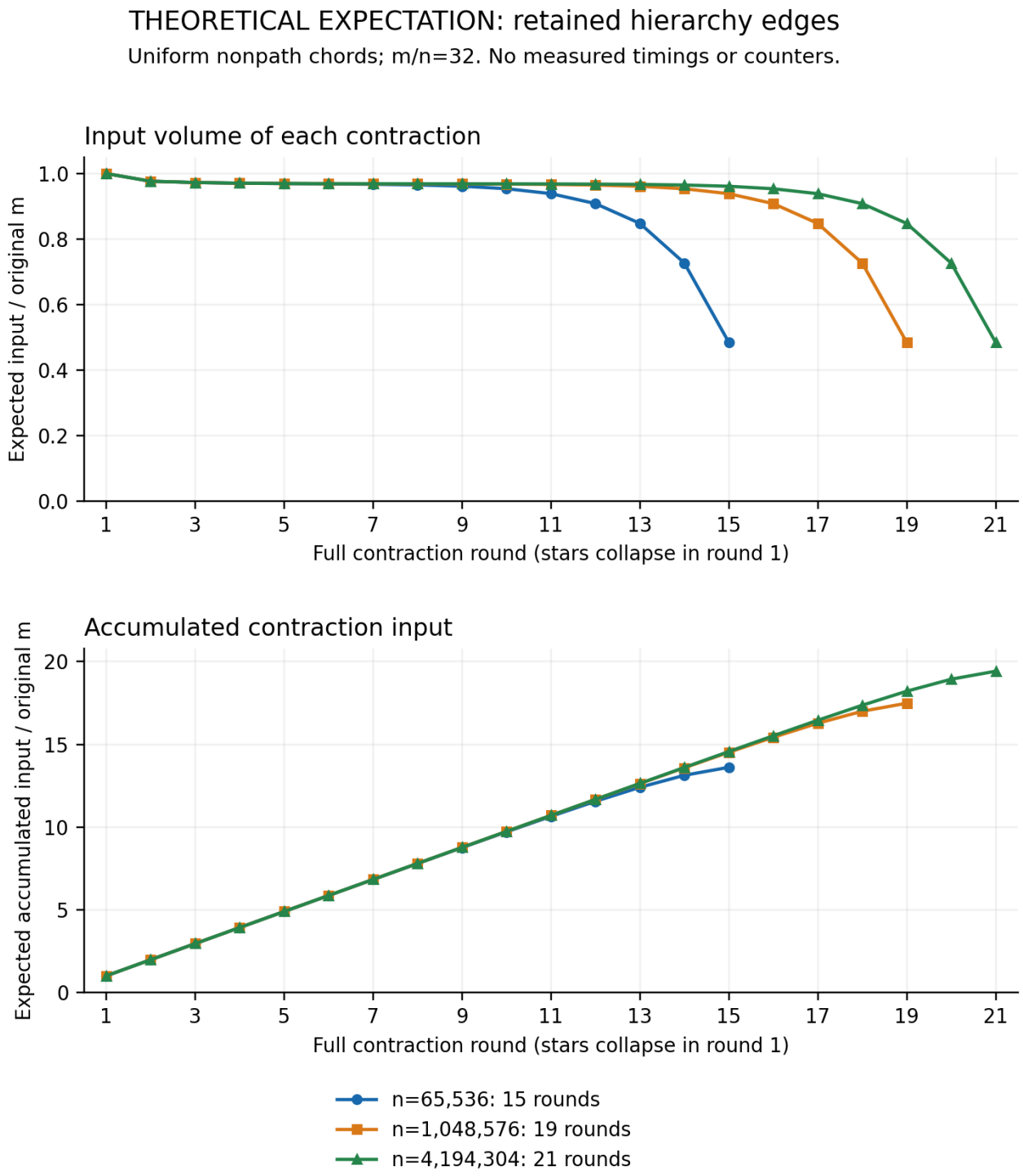
Cheap attachments alone do not promise cheap edge reduction. At m=32n, the first star contraction removes 75% of vertices but only **2.34375% of edges**. The next hierarchy contraction pairs core vertices. Together the two rounds remove 87.5% of vertices while retaining **97.265625% of edge**

records. A large, dense residual problem remains even though its component arrays have become small.

After k core rounds, blocks contain b=2^k consecutive core vertices. A chord remains unless both ends are in the same block. There are D=choose(c,2)−(c−1) eligible nonpath pairs, and the number internal to those blocks is

$$I_b = (c / b) \cdot \text{choose}(b, 2) - (c - c / b).$$

Subtracting c−c/b excludes path pairs already internal to blocks. If C=m−n+1 distinct chords are chosen uniformly, expected surviving chords are C(1−I_b/D); the remaining path contributes c/b−1 edges. Early blocks contain only a small fraction of possible pairs, so almost all chords continue crossing them.



Theoretical expected retained edges and cumulative input volume; not measured timings or counters

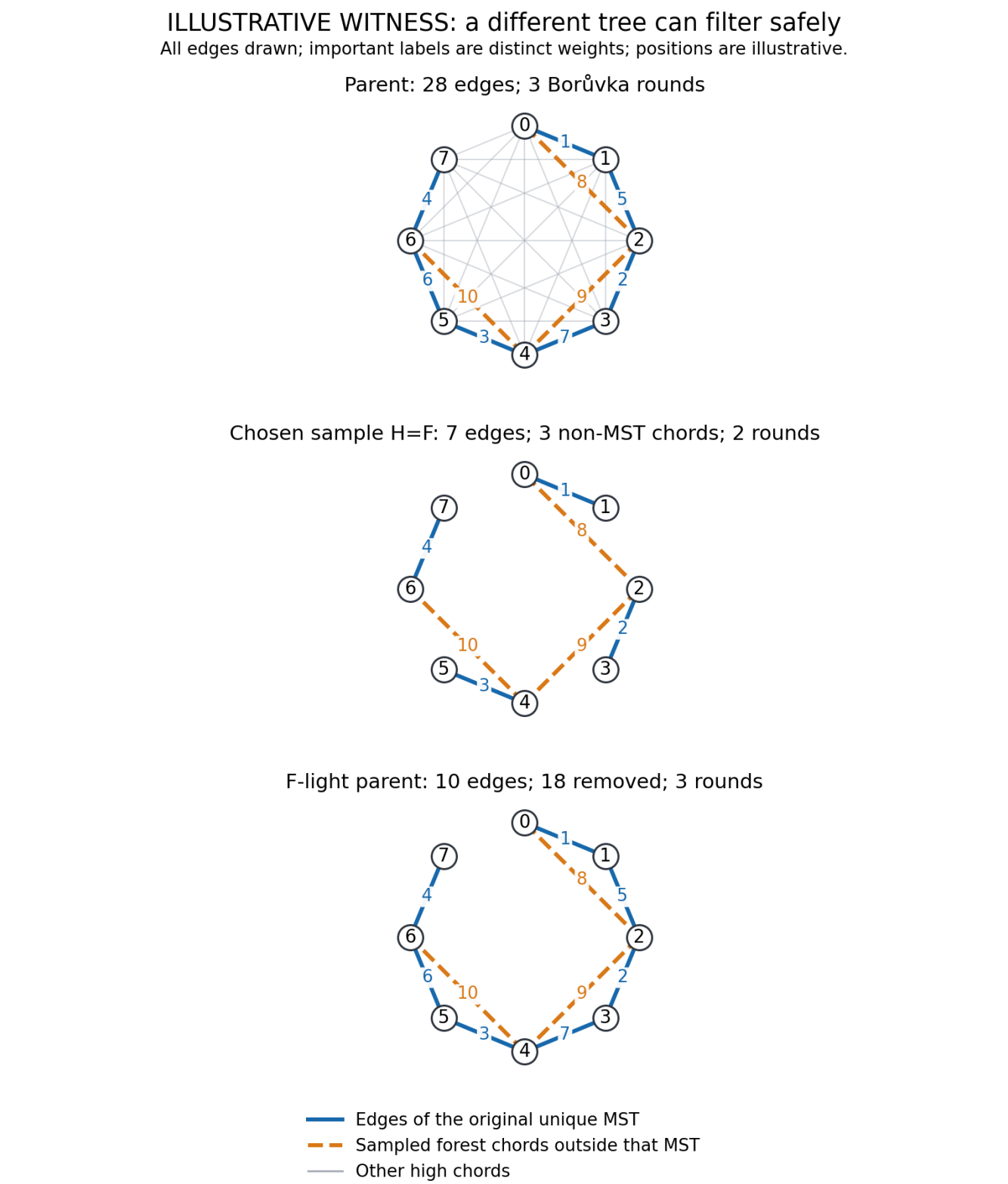
Including the initial star input and core inputs b=1,2,4,…,c/2 gives expected contraction-input volumes **13.6115m, 17.4846m and 19.4219m** for the three larger-study sizes, over 15, 19 and 21 full contractions. This is an expectation under ideal uniform distinct-chord sampling. Tiny independent enumeration checked the combinatorial formula before rendering. It is not a same-seed work trace from the large performance campaign.

For the growing ideal uniform-chord family at fixed density d=32 (or any fixed d>1), C=Θ(m), so expected retained-contraction input volume is Θ(m log n). This is an ensemble expectation, not a per-realization Θ claim. The implementation and finite generator have bounded integer/memory domains; the formula does not predict a feasible crossover. Increasing density at fixed n pushes C/m toward one but adds no rounds. Increasing n adds levels. This explains why multiplying density can have a limited effect on the particular repeated-scan opportunity.

There is an eventual advantage in a precisely limited mathematical sense. For this ideal, unbounded family, if KKT's linear primitives and fresh independent sampling at fixed p hold uniformly, its expected RAM work is O(m). Charging both algorithms fixed positive primitive costs gives E[W_KKT]/E[W_retained B]=O(1/log n)→0. This is a ratio of expectations, not an expected per-input ratio or a guarantee for each seed. It predicts no finite clock threshold, advantage over every solver, or asymptotic theorem for the bounded seeded executable.

The feature that may generalize is persistent crossing records, not many rounds alone. Trees can contract for many levels while their edge volume falls geometrically; cheap fixed-size modules can similarly discard most records early. Hierarchical costs help this mechanism only when Θ(m) records keep crossing many component levels. Community labels or geometric drawings do not ensure that condition, and the source's community_distance4 endpoint redistribution has the same core quotient, not an independent mechanism. The topology-identical shuffled losses are controlled evidence that weight organization matters, not a theorem about all random weights.

KKT can remove heavy chords without waiting for their endpoints to enter one Borůvka component. After its first two hierarchy contractions at density 32, the root has n/8 vertices and 249m/256 edges. Half sampling gives a sample branch of expected size 249m/512, while the later light branch has expected size at most m/128. The latter small bound is attractive, but KKT must first solve the much larger sample and construct its verifier.



A checked eight-vertex witness of sampling and filtering; not a timing fixture or typical draw

The witness contains 28 edges. Its chosen seven-edge sample forest contains three edges absent from the true MST, yet it certifies 18 heavy removals and leaves ten light edges containing the MST. Independent Kruskal, path and Borůvka checks produce full/sample/filtered round counts **3/2/3**. The

sample can connect across hierarchy levels through edges that will not survive the final MST. That is why “the sample contains non-MST edges” does not make its filtering useless.

An edge-volume count is still not runtime. On this loop-free connected hierarchy, original Borůvka performs one full original-edge loop per productive round plus a final no-op loop: **16m, 20m and 22m** edge iterations at the three sizes. For retained contraction with summed input volume V and terminal input t, the helper performs **3V−2t** edge-loop iterations and emits V−m quotient records, besides its initial conversion. Vertex work, hashing, allocation, branches and memory accesses differ. The baseline source accounting explains the distinction without assigning measured CPU costs to those operations.

9. Common families: historical successes and current limits

The earlier common-family study did find practical KKT successes against its three classical repository baselines. The repaired production source led all three by case medians on 22 of 52 labels in the primary run and 21 in the repeat. The same 21 labels led on every graph median in both runs. These labels include a deliberate fixed-256/square-root small-side alias at n=65,536, producing three duplicated weighted inputs. They are not 21 independent topology discoveries.

Small vertex covers explain the strongest historical cases. If every outside vertex connects to a cover of size k, each selected component after one Borůvka round contains a cover vertex; there are at most k components. Deduplication then bounds the residual by choose(k,2) edges. With fixed k, the recursive tail stays small while input/output work grows. At n=262,144, fixed-16 bipartite KKT took 28.542 ms versus 61.995 ms Kruskal, 290.424 ms Prim and 55.557 ms original Borůvka in the primary campaign.

That mechanism differs from cheap degree-one attachments. The bipartite examples have no leaves. Ninety-percent designated attachments offered a modest repeated Kruskal lead at larger sizes, whereas fifty-percent attachments did not. Rings of small cliques could contract most edges, but a substantial near-tree residual remained and the advantage disappeared at the largest historical point. In that campaign, every measured sparse grid variant and the regular/random controls lost to Kruskal. A high degree, many leaves or a small retained-edge percentage is not a sufficient classifier.

The historical continuation ablation favored compacting Borůvka in 18 of 20 cases. Thus even those successful four-way results did not establish that sampling was the source of the advantage. The later retained-edge control strengthens this caution. Historical wins remain valid for their competitors and revisions, but do not fill missing columns in a current six-way leaderboard.

The newer held-out screen uses compressed queries and **scalar half sampling**: six families, six sizes from 1,024 to 1,048,576, density 32, two fresh graph seeds and one sampler. “Held out” means the prospective matrix and seeds, not previously unknown family definitions. Its 72 children produced 3,024 certified outputs. All thirty family/comparator outcomes are retained:

Family	vs dedup B	vs retained B	vs Kruskal	vs Prim	vs original B
Leaf distance	Strong bracket (16,384,262,144]	Narrow only at 1M	Loss throughout	Loss throughout	Loss throughout
Shuffled core	Same strong bracket	Loss throughout	Small lead reverses	Loss throughout	Loss throughout
Grid distance ranks	Same strong bracket	Loss throughout	Mixed at 4K, otherwise loss	Strong from 16K; no earlier strict loss	Loss throughout
Grid random ranks	Same strong bracket	Loss throughout	Narrow through 16K, then loss	Strong bracket (4,096,65,536]	Loss throughout

Family	vs dedup B	vs retained B	vs Kruskal	vs Prim	vs original B
Line distance ranks	Same strong bracket	Loss throughout	Loss throughout	Strong at every size	Loss throughout
Line random ranks	Same strong bracket	Loss throughout	Small leads/mixed reverse	Strong bracket (4,096,65,536]	Loss throughout

There are eight strong brackets: six against deduplicating Borůvka and two against Prim. None is against retained-edge Borůvka, Kruskal or original Borůvka. Original Borůvka wins **all 432 direct trial pairs on both clocks**, ruling out an all-six winner. The near-parity leaf-distance endpoint wins only seven of twelve retained-B pairs; its four graph/clock medians span 0.99355-0.99887.

At n=1,048,576, the complete five-way ratios are below. Every cell is wall/CPU. Equal rounded ratios do not imply equal raw durations.

Family	/ dedup B	/ retained B	/ Kruskal	/ Prim	/ original B
Leaf distance	.3307/.3307	.9960/.9959	1.5426/1.5426	1.3348/1.3348	1.0336/1.0336
Shuffled core	.3433/.3433	1.5772/1.5772	1.3238/1.3238	1.5920/1.5920	1.5550/1.5550
Grid distance	.3691/.3691	1.4106/1.4106	1.4791/1.4791	.4405/.4405	1.1762/1.1762
Grid random	.3589/.3589	1.5645/1.5645	1.3327/1.3327	.5553/.5553	1.4406/1.4406
Line distance	.3581/.3581	1.6387/1.6387	1.5460/1.5460	.1723/.1723	1.2717/1.2717
Line random	.3600/.3600	1.5402/1.5402	1.3380/1.3380	.5389/.5389	1.4166/1.4166

The strong line-distance result against Prim is real, but other solvers remain faster. These line/grid fixtures have path or jittered-grid scaffolds plus shortcuts. They are not road networks, k-nearest-neighbor graphs or pure planar meshes. Distinct distance ranks need not satisfy a triangle inequality. At the smallest attachment size the core is 50/51 complete, whereas larger cores maintain average degree 250 with much smaller occupancy.

Line gaps come from 1…1,024 with a fixed tie order. Full-graph contractions remain intervals, and each round removes all boundaries of the smallest remaining gap. That finite alphabet bounds their rounds by 1,024 as well as by the usual halving bound. It does not apply to arbitrary recursive sampled subgraphs. A geometry label alone therefore supplies no unlimited logarithmic separation.

The completed untimed follow-up checks the same 72 fingerprints, 216 outputs and 144 full comparator replay chains. At 1M, ordered attachments require 19 retained-B rounds and 51.49 contraction-edge visits per original edge; shuffled twins require 8-9 rounds and 21.20-22.66 visits. KKT's corresponding visits are 11.65 and 10.74-10.99. Grid-distance needs eleven comparator rounds, line-distance nine. These counts support a repeated-contraction mechanism but cannot convert heterogeneous work into CPU time.

Existing small graph illustrations help interpret construction, with their own explicit sizes, seeds and older contraction stages. They are not downsampled million-vertex inputs or traces of the latest binary. A historical browser receipt records 70 stage checks, desktop/mobile controls and offline behavior passing for identified old edition hashes. That receipt applies to those historical hashes; it does not validate the current viewer or this edition.

The separately generated current root-stage viewer uses the exact ordered/shuffled n=1,024, m=32,768 fixtures with graph seed 17000011 and sampler seed 2801. Its export, independent graph checks and timing-input join passed. It reconstructs two contractions, half/eighth root samples and heavy-edge filtering; it is neither a full recursion trace nor a half/eighth timing comparison. Complete data are embedded while background drawing is capped at 512 edges, with essential forest edges retained. The earlier readiness receipt and fresh browser-policy receipt leave desktop/mobile interaction and visual QA incomplete. The new CUA attempt found no in-app browser; Chrome

explicitly rejected the local file URL before loading and prohibited workarounds. No localhost server, republishing or alternate-surface bypass was attempted. This is a delivery restriction, not a rendered-viewer failure or interactive-validation pass.

10. Half sampling at four million vertices: one crossover and one reversal

The larger study reuses compressed-query scalar-half KKT at n=65,536, 1,048,576 and 4,194,304, m=32n. Two twins/two fresh graphs give twelve children, 504 certified outputs and 432 timed calls. Independent audit agrees on 720 clock ratios, 396 summaries, thirty point states, thirty endpoint pairs and ten outcomes. The complete table and wall plot retain all comparators and sizes.

Family / n	/ dedup B, wall/CPU	/ retained B	/ Kruskal	/ Prim	/ original B
Hierarchy 65,536	.8077/.8077	1.1181/1.1181	1.2702/1.2702	2.2681/2.2681	1.6442/1.6442
Hierarchy 1,048,576	.3191/.3191	.9674/.9674	1.5101/1.5101	1.7631/1.7630	1.0111/1.0111
Hierarchy 4,194,304	.3924/.3923	1.0786/1.0784	2.1109/2.1106	1.6817/1.6816	.8861/.8860
Shuffled 65,536	.8607/.8607	1.7382/1.7382	1.1059/1.1059	1.9909/1.9909	2.5197/2.5197
Shuffled 1,048,576	.3386/.3386	1.5199/1.5199	1.3048/1.3047	1.5310/1.5309	1.5613/1.5612
Shuffled 4,194,304	.3994/.3991	1.5337/1.5342	1.6799/1.6796	1.3598/1.3586	1.2785/1.2778

Hierarchy/retained B is **loss → narrow lead → loss**: eleven of twelve 1M trial wins, medians 0.96375-0.97358, become zero wins at 4M, medians 1.07229-1.08749. Hierarchy/original B is **loss → loss → strong lead**: all four medians exceed one at 1M and reach 0.88254-0.88614 at 4M, establishing **(1,048,576,4,194,304]**. All twelve largest trials win, not all by 10%; no larger observation or interpolated transition is claimed.

The other eight outcomes remain: both twins strongly beat dedup B throughout; shuffled retained/original B and both twins' Kruskal/Prim comparisons lose throughout. Across 72 pairs per comparator/clock, wins are 72/11/0/0/15 against dedup B/retained B/Kruskal/Prim/original B. There is no all-six winner.

Half-policy CPU comparisons preserve the complete second clock, including the retained-B reversal.

At 4M and **134,217,728 edges**, ordered median wall times are **21.818 s KKT, 55.312 s dedup B, 20.249 s retained B, 10.313 s Kruskal, 12.894 s Prim and 24.604 s original B**. Shuffled KKT takes 17.232 s, versus 11.325 s retained B and 10.236 s Kruskal. Absolute time medians provide scale; they are not substitutes for the paired-ratio statistic.

Half-policy absolute wall times retain all six solvers.

The earlier direct-Edge density screen reached 67 million edges at density 64, yet retained B won 96/96 pairs on both clocks: ordered KKT was 4.2-12.0% slower, shuffled 73.7-88.6% slower. Different verifier/seeds preclude pooling; density alone did not deliver a win.

Half-policy absolute CPU times use matching wall/CPU panel bounds.

All six larger-study figures, including the two analytic illustrations above, have original-resolution visual QA from two reviewers. The theoretical retention curves are explicitly not measured counters from these 4M timing children.

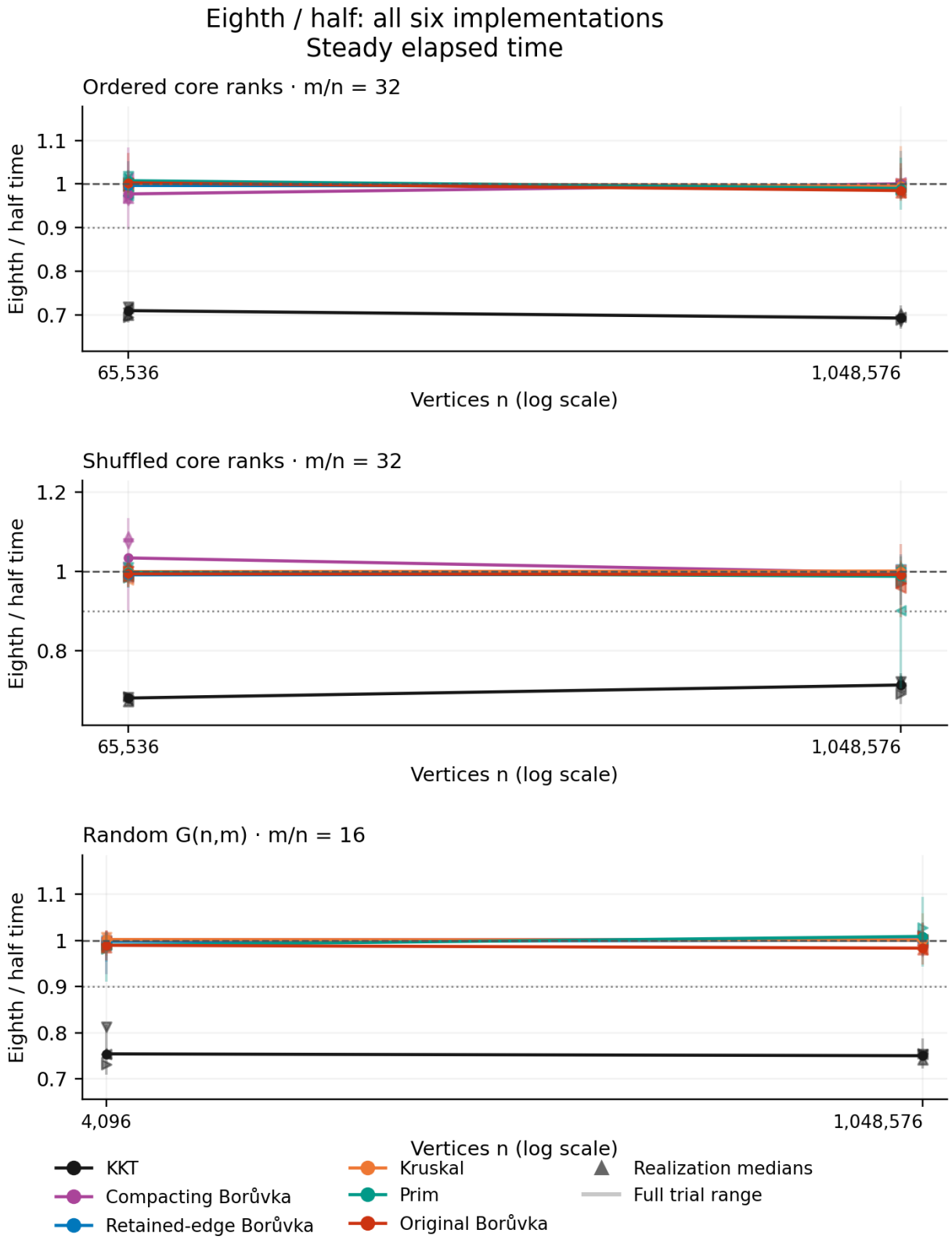
11. Eighth sampling: a substantial retained-B lead, with clear limits

The completed rate experiment changes only the private scalar predicate: half accepts `(word & 1) == 1`, while eighth accepts `(word & 7) == 7`. Both consume one MT19937 word per visited edge. Both retain reserve(m/2), compressed verifier queries, direct Edge input and parallel quotient records. The public sampler remains half sampling, and the ordinary build is unchanged. Only the KKT object differs; seven production objects and the timing main are physically shared. This is an **opt-in implementation result**, separate from packed-bit sampling.

Fresh half and eighth binaries ran the same six cases, two graph seeds and two sampler seeds, with six timed repetitions per realization. Every one of the **144 paired KKT trials improved on both clocks**. The six case medians use **24.6-32.0% less time**; no historical timing is pooled into this comparison.

Case	Eighth/half wall	Process CPU
Ordered 65,536	0.70977	0.70977
Ordered 1,048,576	0.69269	0.69270
Shuffled 65,536	0.68019	0.68020
Shuffled 1,048,576	0.71311	0.71311
G(n,m) 4,096	0.75383	0.75380
G(n,m) 1,048,576	0.74988	0.74988

Unchanged-control case ratios span **0.97735-1.03339** across both clocks. Their drift is displayed without correction or subtraction. Four graph/sampler realizations are not 24 independent graphs, repeated seeds reproduce a realization, and wall/CPU agreement is not an independent replication.



24 trials per point; four graph/sampler medians. Complete ranges, not confidence intervals.
No normalization; lines connect observations only; every clock flag retained.

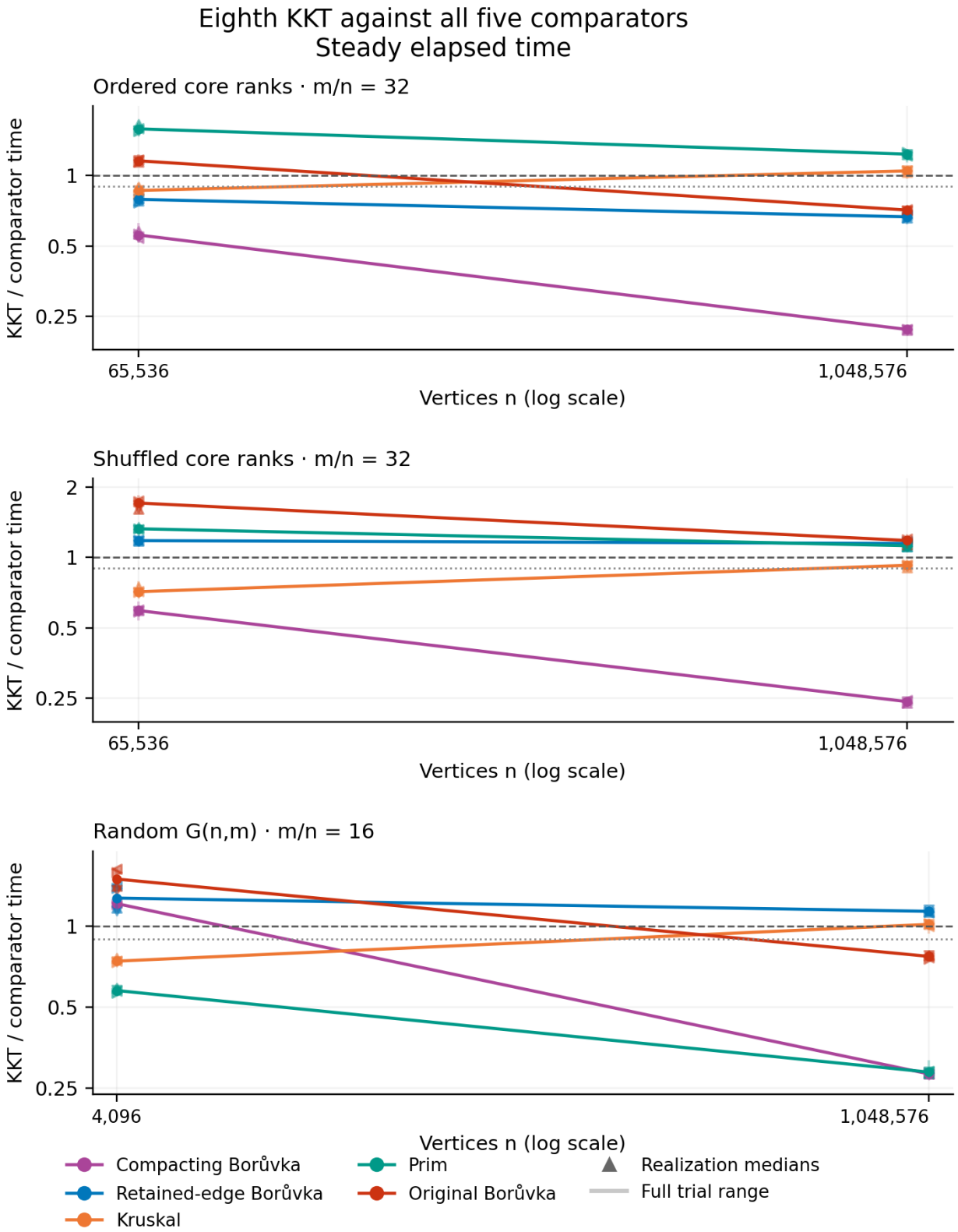
Fresh eighth/half ratios for all six algorithms, including unchanged controls

The full direct comparison is more informative than the implementation gain alone. Each cell below is **wall/CPU**, the median of 24 paired KKT/comparator ratios. Values below one favor eighth KKT. Attachment cases use m/n=32; G(n,m) uses 16.

Eighth case	Dedup B	Retained B	Kruskal	Prim	Original B
Ordered 65,536	0.5564/0.5564	**0.7891/0.7891**	0.8612/0.8613	1.5769/1.5769	1.1523/1.1523
Ordered 1,048,576	0.2200/0.2200	**0.6648/0.6648**	1.0435/1.0435	1.2301/1.2301	0.7103/0.7103
Shuffled 65,536	0.5930/0.5930	1.1794/1.1794	0.7138/0.7138	1.3254/1.3254	1.7076/1.7076
Shuffled 1,048,576	0.2420/0.2421	1.1472/1.1472	0.9248/0.9248	1.1216/1.1216	1.1808/1.1808
G(n,m) 4,096	1.2157/1.2150	1.2748/1.2745	0.7432/0.7429	0.5780/0.5780	1.5003/1.4993
G(n,m) 1,048,576	0.2832/0.2832	1.1384/1.1384	1.0182/1.0182	0.2875/0.2875	0.7740/0.7740

The ordered retained-B lead is substantial: about **21.1% less time at 65K and 33.5% at 1M**. All 24 trial ratios at each size are below one on each clock. At 1M, wall realization medians range from 0.66118 to 0.67697, and even the largest individual wall ratio is 0.69830. However, both endpoints already strongly lead. Within that paired study there is no smaller observed loss from which to claim a size crossover. The separate completed extension below adds sizes through 4M without pooling the campaigns.

All shuffled and G(n,m) trials still lose to retained B. Across the complete 144 pairs per comparator and clock, eighth KKT wins 120 against dedup B, 48 against retained B, 101 against Kruskal, 48 against Prim and 48 against original B. On ordered 1M, Kruskal wins 22/24 and Prim all 24. The fastest available solver remains family-dependent; there is no all-six winner.



24 trials per point; four graph/sampler medians. Complete ranges, not confidence intervals.
No normalization; lines connect observations only; every clock flag retained.

Eighth KKT against every comparator; all losses and both measured endpoints remain

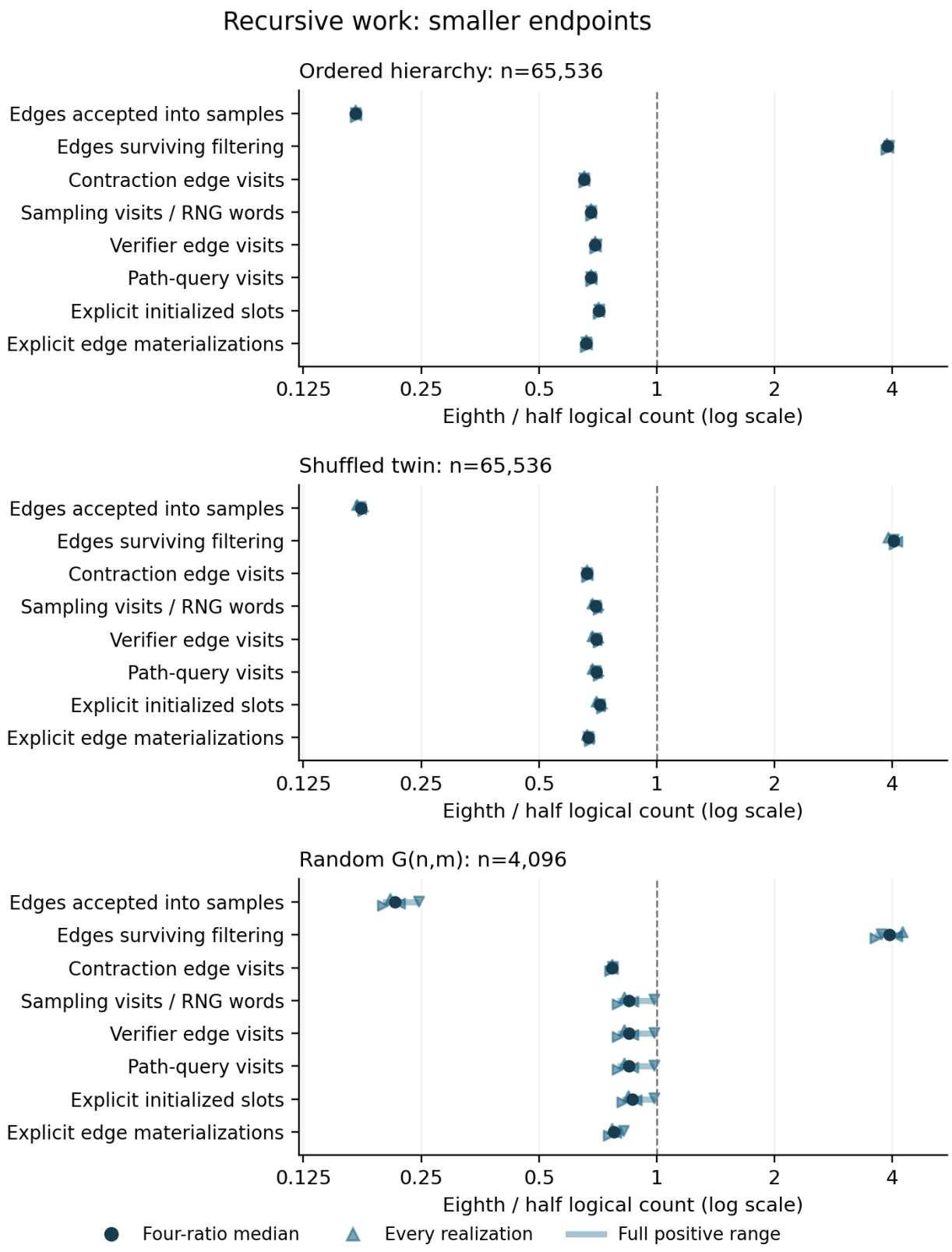
At ordered 1M and 33,554,432 edges, median wall times are **2.402 s KKT, 3.623 s retained B, 10.946 s dedup B, 2.298 s Kruskal, 1.957 s Prim and 3.398 s original B**. Dividing these absolute medians does not reproduce a median of paired ratios. The complete both-clock tables retain both policies and every realization/range; the 30 classifications retain reversals as well as successes. Eighth establishes loss-to-strong-lead brackets against original B on hierarchy and G(n,m), and against

dedup B on G(n,m). Its earlier Kruskal leads instead become losses at the large hierarchy and G(n,m) endpoints.

The source tradeoff helps explain why testing p was worthwhile. After the ordered 1M graph's first two contractions, q=131,072 and r=32,636,928. Changing p from 1/2 to 1/8 reduces expected sample edges from **16,318,464 to 4,079,616**, while increasing the expected-light-edge bound from **262,144 to 1,048,576**. Those are theoretical expectations/bounds, not measured branch counts or a CPU breakdown. Recursive forests and RNG positions can differ despite equal initial seed numbers.

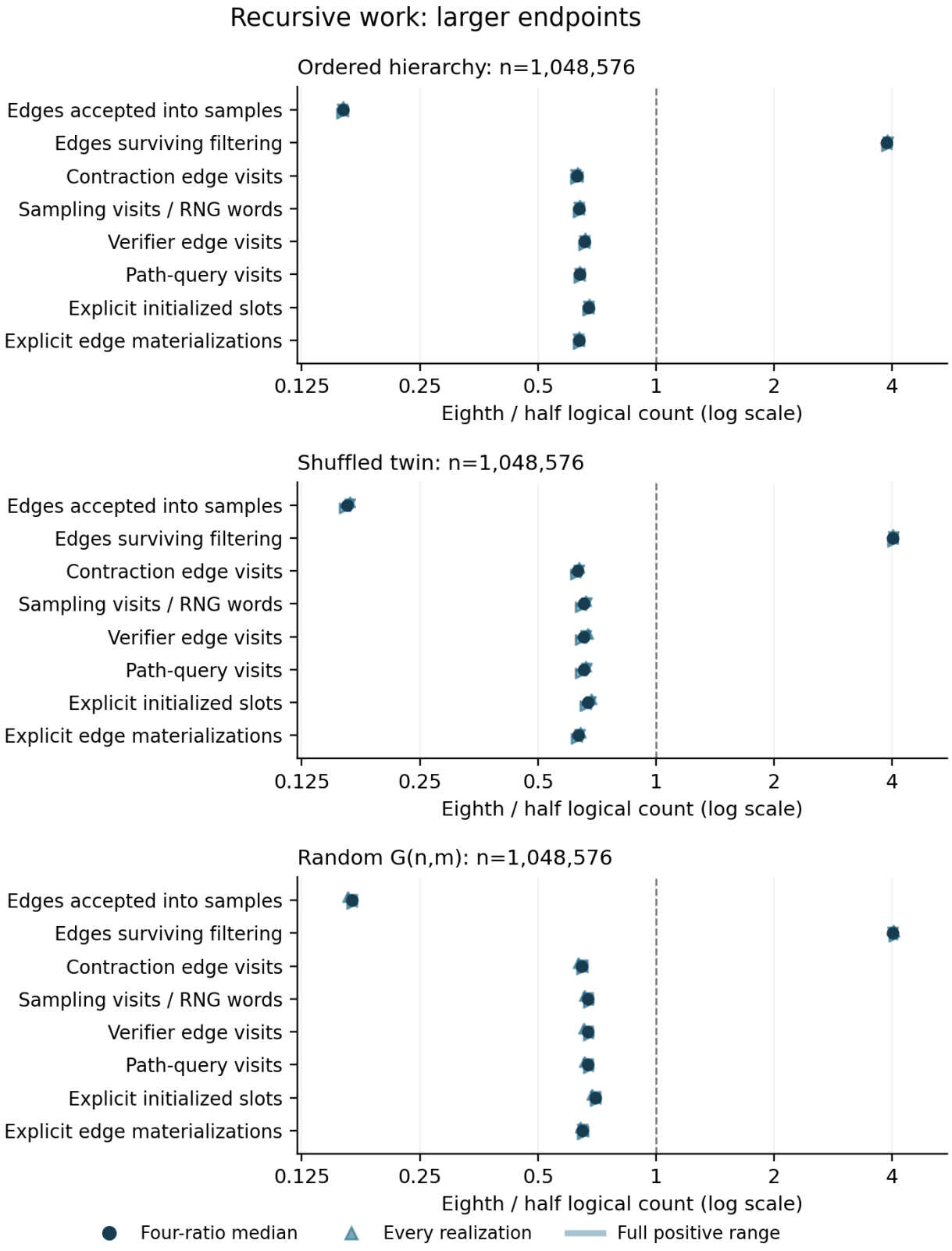
The separate completed work study now checks all 48 corresponding instrumented inputs, 144 certified outputs, 96 comparator replay chains and 1,000 round rows. An independent audit reconstructs all 9,360 counter cells and 1,608 ratio records, retaining undefined both-zero cases. It confirms unchanged comparator continuations and initial KKT contraction dimensions, while allowing the later recursion to change. The independent mechanism review supports a narrower conclusion than a CPU profile: a smaller sample branch can outweigh more surviving edges in several important kinds of recursive work.

For ordered 1M, median paired eighth/half ratios are **0.1591 accepted sample edges, 0.6364 sampling visits, 0.6281 contraction edge visits, 0.6738 initialized slots and 0.6356 explicit edge materializations**. Cumulative filtered survivors instead grow **3.8808×**. Both policies still draw one scalar RNG word per visited sampling edge; the lower word total comes from fewer later visits, not fewer words during a given scan. Accepted-edge totals fall by more than fourfold because both acceptance probability and cumulative sampling volume fall. These are four paired realization ratios per case, with two graph seeds and two sampler seeds, not four independent graphs.



All recursive calls; two graphs x two sampler seeds. Logical counts are not CPU costs.

Separately audited recursive work at all three smaller endpoints: four realization ratios and full ranges



All recursive calls; two graphs x two sampler seeds. Logical counts are not CPU costs.

The same eight preselected fields at all three larger endpoints, including the growing filter branch

The two figures retain all six cases and eight fields selected before diagnostics. Every realization reduces its seven displayed non-filter fields and increases filtered survivors. Small G(n,m) sampling-visit ratios range from **0.7936 to 0.9840**, so savings are not uniformly large. Across all 1,560 paired fields, **774 decrease, 491 increase and 295 are equal**; 59 ratios are both-zero and

remain explicit. These fields mix volumes, calls, depths and operations and cannot be added into a runtime score. The 96 unmeasured comparator hash-insertion cells remain null.

Survivors sum over recursive verifier calls; they are neither root survival probabilities nor disjoint edge sets. Query sharing also changes: ordered 1M's per-solve cumulative U/Q spans **0.7908-0.8959% for half** and **1.2792-1.4944% for eighth**, while total path-query visits still fall by a median 36.2%. These are ratios of cumulative per-call counts, not a distribution of individual calls or global uniqueness. The measured work supports the timing improvement without isolating cache misses, allocator effects or CPU attribution. Retained B still wins the shuffled and random timing cases.

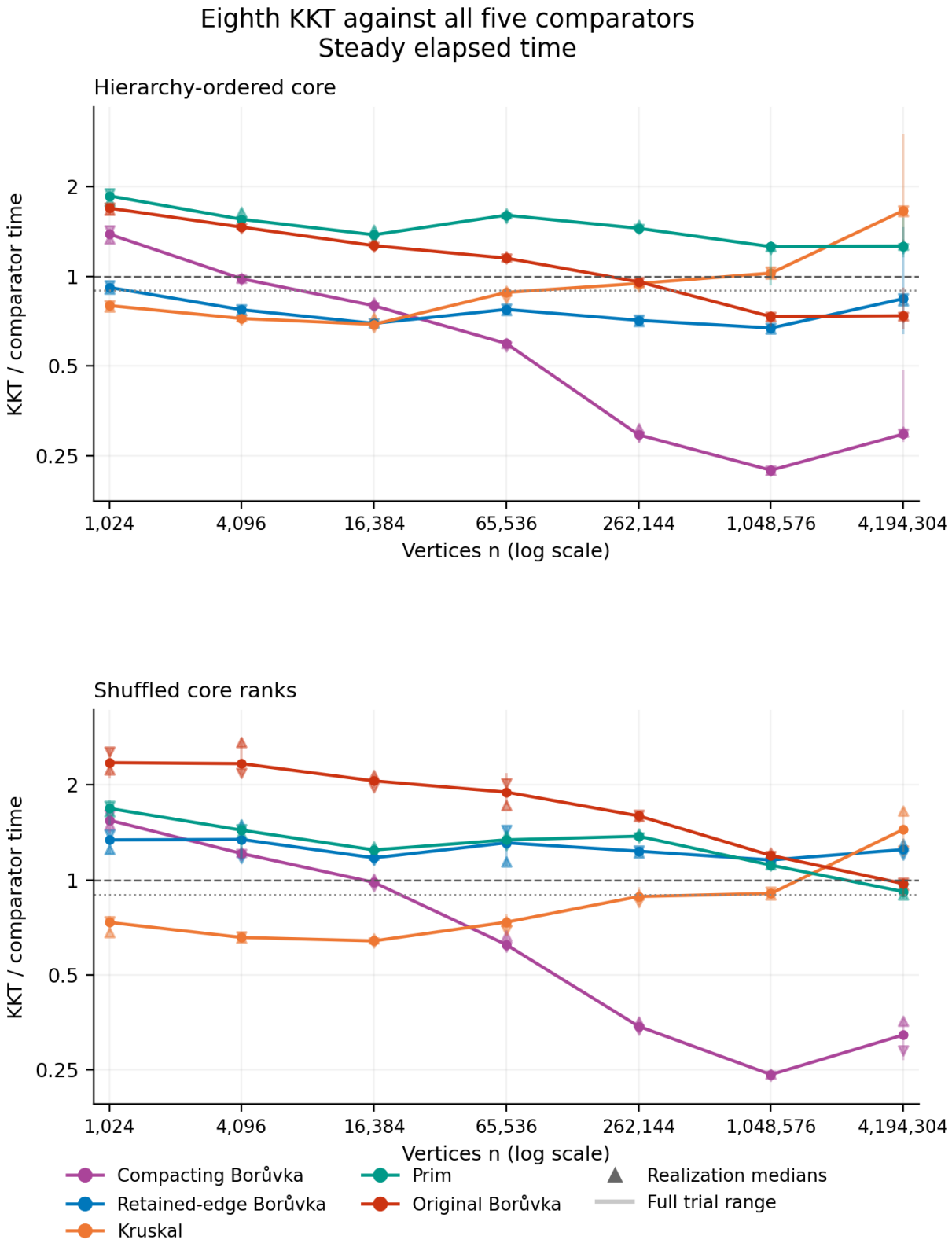
All 2,016 solver outputs passed validation. Strict analysis and the independent exact-arithmetic raw audit agree on 4,608 ratios, 3,024 summary rows and all 30 classifications. Native and ASan/UBSan profiles each passed 20 positive suites plus four wrong-policy negatives; full-solver gates preserve 1,344 input/MSF pairs, and the fresh half path replays the parent's 65 counters. Final linked-object checks are separate. All ten standalone timing figures passed original-resolution visual QA; two of those timing figures are embedded here, alongside the two separate work figures.

The completed seven-size extension: useful, but not universal

The prospectively declared size ladder reuses the exact eighth binary with all five controls. Its fourteen cases pair ordered and shuffled core weights at seven sizes from 1,024 to 4,194,304, m=32n, two fresh graph seeds and one sampler. The declaration preceded inspection of the paired eighth results. All 28 children passed: 1,176 certified outputs, including 1,008 timed calls and 168 warmups. An independent raw audit agrees on all 1,680 ratios, 924 summary rows, seventy point states, 210 endpoint pairs and ten outcomes. This is a new size campaign, not a half/eighth implementation comparison.

n	Ordered KKT/retained B wall/CPU	Shuffled KKT/retained B wall/CPU
1,024	0.9179/0.9195	1.3398/1.3392
4,096	0.7730/0.7727	1.3444/1.3441
16,384	0.6956/0.6955	1.1752/1.1757
65,536	0.7750/0.7750	1.3116/1.3117
262,144	0.7124/0.7124	1.2340/1.2340
1,048,576	0.6713/0.6713	1.1571/1.1571
4,194,304	0.8414/0.8413	1.2482/1.2481

The ordered case is a **narrow lead at 1,024 and a strong lead at all six larger points**, on both graphs and clocks. There is no smaller measured loss, hence no retained-B crossover bracket under the declared definition. Every ordered trial through 1M wins. At 4M, however, only **9/12** trials win per clock: wall ratios range **0.64095-1.46180**, CPU **0.64862-1.44673**. Its four graph/clock medians remain 0.83103-0.84144. The strong median classification and the three individual losses must be stated together. All 84 shuffled retained-B pairs lose on both clocks.



12 trials per point; two graph medians. Complete ranges, not confidence intervals.
No normalization; lines connect observations only; every clock flag retained.

Seven-size eighth KKT comparisons: all five controls, both weight twins and every trial-range endpoint

The complete ten family/comparator conclusions are:

Comparator	Ordered hierarchy	Shuffled core ranks
Dedup B	Strong sustained bracket **(1,024,16,384]**	Strong sustained bracket **(4,096,65,536]**
Retained B	Narrow then strong leads; no measured lower loss	Loss at every size

Comparator	Ordered hierarchy	Shuffled core ranks
Kruskal	Early strong leads reverse to losses at 1M and 4M	Early strong leads reverse to a loss at 4M
Prim	Loss at every size by graph/clock medians	Loss through 1M; narrow lead at 4M
Original B	Strong sustained bracket **(65,536,1,048,576]**	Loss through 1M; narrow lead at 4M

Each bracket requires a smaller loss and a later at-least-10% lead on both graphs/clocks, maintained at every larger measured size. It supplies a measured interval, not an interpolated crossing or a promise beyond 4M. The other seven outcomes are not hidden. At ordered 4M, wall/CPU ratios against dedup B, Kruskal, Prim and original B are **.2960/.2969, 1.6616/1.6618, 1.2638/1.2640 and .7379/.7379**. The full both-clock table and ranges retain all fourteen cases.

KKT beats Prim in fourteen of 168 pairs per clock: two ordered-1M pairs and all twelve shuffled-4M pairs. One ordered-1M wall repetition beats all five controls simultaneously; no CPU repetition does. There is no case-level all-six winner by the declared medians. Small-case wins against Kruskal reverse as size grows, and the retained-B margin is nonmonotonic. These details rule out a general “larger means KKT dominates” claim.

The full extension's absolute wall times and absolute CPU times retain all six solvers in the supplementary gallery.

At ordered 4M, wall medians are **17.752 s KKT, 57.310 s dedup B, 19.683 s retained B, 10.416 s Kruskal, 14.116 s Prim and 24.211 s original B**. KKT's wall range is 15.153-36.876 s; no slow call is trimmed. These are absolute medians, not the inputs to a ratio-of-medians substitution. Both CPU figures and all exact data remain linked in the study; all four standalone PNGs passed visual inspection. Page-scale QA remains separate.

This extension measures timing, not a new counter breakdown. The earlier retention theorem and paired work results provide qualified structural explanations, not substituted per-input measurements. The nearly complete 256-vertex core at the smallest size becomes proportionally sparser along the ladder despite fixed average core degree 250. The largest input reaches the existing n and m limits exactly. Two graph seeds and one repeated sampler, reversed schedules and a finite machine leave substantial limits on generalization.

Distance-ranked cores: another useful weight arrangement

The completed distance study retains the same eighth binary and all five controls, with fresh graph seeds 21000011/21000041 and sampler 3203. Twelve cases cover distance/shuffled twins at six sizes, 1,024 through 1,048,576, with m=32n. All 24 children and 1,008 outputs passed; the independent audit agrees on 1,440 paired ratios, 792 summary rows and all ten outcomes. No half-policy timing or historical control is pooled.

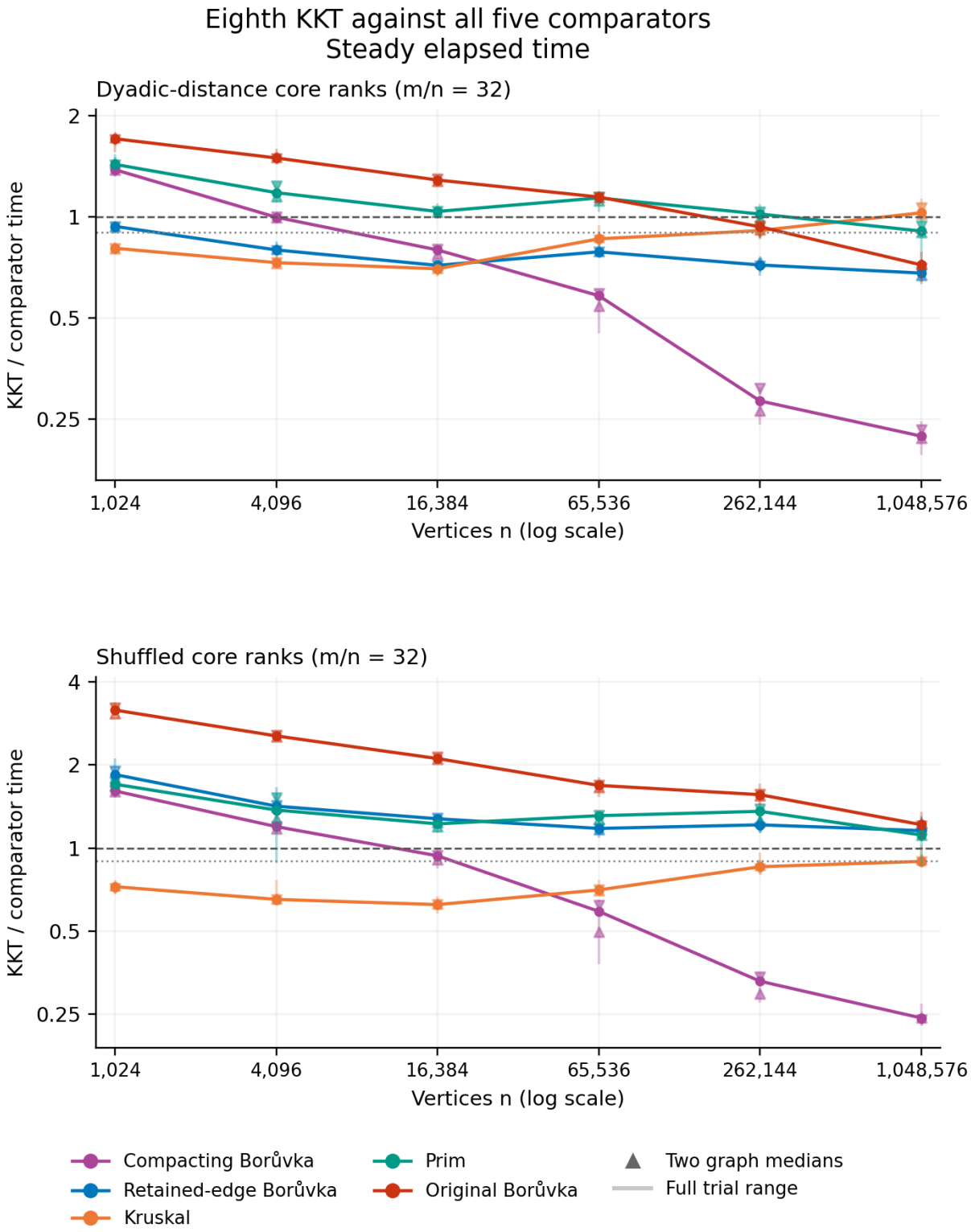
n	Distance KKT / retained B, wall / CPU	Shuffled KKT / retained B, wall / CPU
1,024	.937703/.937932	1.847185/1.846045
4,096	.797872/.797999	1.418369/1.418314
16,384	.717690/.719921	1.276525/1.276478
65,536	.787462/.787712	1.177434/1.177430
262,144	.719496/.720126	1.213465/1.214256
1,048,576	.680665/.680418	1.156119/1.156126

All 72 distance trial pairs beat retained B on each clock; all 72 shuffled pairs lose. The five larger distance points meet the strong criterion on both graphs/clocks, but the smallest already narrowly leads. Thus the second family also supplies a practical advantage without an observed lower losing

endpoint. At 1M, the distance wall trial range is .630696-.745135 and the CPU range .632049-.743753. The four graph/clock medians span .671389-.684276. The margin is not monotonic across sizes.

Three strong brackets are observed: distance versus compacting B **(1,024,16,384]**, shuffled versus compacting B **(4,096,65,536]**, and distance versus original B **(65,536,1,048,576]**. The last has no larger measured point. The other seven outcomes remain: distance retained B narrow then strong; distance Kruskal strong through 65K, narrow at 262K, then a 1M loss; distance Prim loss through 65K, mixed at 262K and narrow at 1M; shuffled retained B, Prim and original B lose throughout by graph/clock medians; shuffled Kruskal strongly leads through 262K and narrowly at 1M.

At distance 1M, KKT/comparator wall/CPU ratios are **.2225/.2224** against compacting B, **1.0292/1.0313** against Kruskal, **.9081/.9080** against Prim and **.7200/.7199** against original B. The Prim lead misses the declared 10% threshold. There is no all-six case winner. Five wall repetitions and four CPU repetitions beat all five controls simultaneously, which does not change that case-level conclusion. The full table preserves all twelve cases and both clocks.



Twelve trials per point; two graphs, one fixed sampler. All ranges are descriptive.
Lines connect measured sizes only. All 0 flagged calls retained.
Ratios: dashed 1.00 = parity; dotted 0.90 = strong-lead threshold.

Distance and shuffled core weights: all five comparators and complete trial ranges

Separate wall runtime medians at distance 1M are 2.4776 s for KKT, 3.6089 s for retained B, 2.3509 s for Kruskal and 2.6771 s for Prim. These describe absolute cost; their quotient is not the paired-ratio median above. The family orders all core edges by dyadic ancestry, unlike the hierarchy construction that makes every path edge cheaper than every chord. Some shortcuts can enter its MST. Rank weights need not be metric distances, and these engineered attachments are not road or

nearest-neighbor workloads. This run supplies timing, not new counters or a causal hardware profile. Its earlier half-policy measurements remain separate rather than serving as an implementation-effect estimate.

Lower densities: a sustained retained-B crossover, and a reversal

The completed lower-density study finally supplies a losing-to-winning bracket against the shared retained-edge continuation. **Distance-ranked attachments at m=16n cross from a loss at n=512 to a strong lead by n=8,192, maintained at every measured size through n=2,097,152.** The m=8n ladder is a counterexample to extrapolation: its strong intermediate lead weakens to mixed results at 4M. Shuffled twins lose every retained-B pair at both densities.

The unchanged eighth binary and all six solvers run thirty cases. Density 16 uses n=512, 2,048, 8,192, 32,768, 131,072, 524,288 and 2,097,152; density 8 uses n=256, 1,024, 4,096, 16,384, 65,536, 262,144, 1,048,576 and 4,194,304. Fresh graph seeds 23000003/23000029 and sampler 3407 were declared before this campaign, after the density 32 results. This targeted follow-up is not an unbiased family survey. All 60 children and 2,520 outputs passed, including 2,160 timed calls / 360 warmups. The independent audit reconstructs 3,600 paired clock ratios, 1,980 summaries, 150 point states, 490 endpoint pairs and all 20 outcomes.

Each ladder fixes m=dn and N=(d+1)n. Average core degree is fixed while occupancy falls with size: the smallest cores occupy 122/127 of possible pairs at d16 and 58/63 at d8. Fixed density does not isolate cache, occupancy or other finite-size effects. Both largest inputs have 33,554,432 edges but different vertex counts; they are not equivalent graphs. Distance ranks encode dyadic ancestry, not necessarily geometric or triangle-inequality distances. The shuffled control preserves topology, IDs, labels, tuple order, cheap leaves and the exact weight multiset.

The following complete outcome table uses **L** for loss on all four graph/clock medians, **N** for all four <1 without all four ≤0.90, **S** for all four ≤0.90, and **M** for mixed/tied. The two chronological ladders above define every position. A strong bracket requires an earlier L, a later S, and S at every remaining measured size; exact fractions decide the thresholds. The representative bracket uses the earliest qualifying upper point and nearest earlier loss. It is not an interpolated threshold or evidence beyond the last observation.

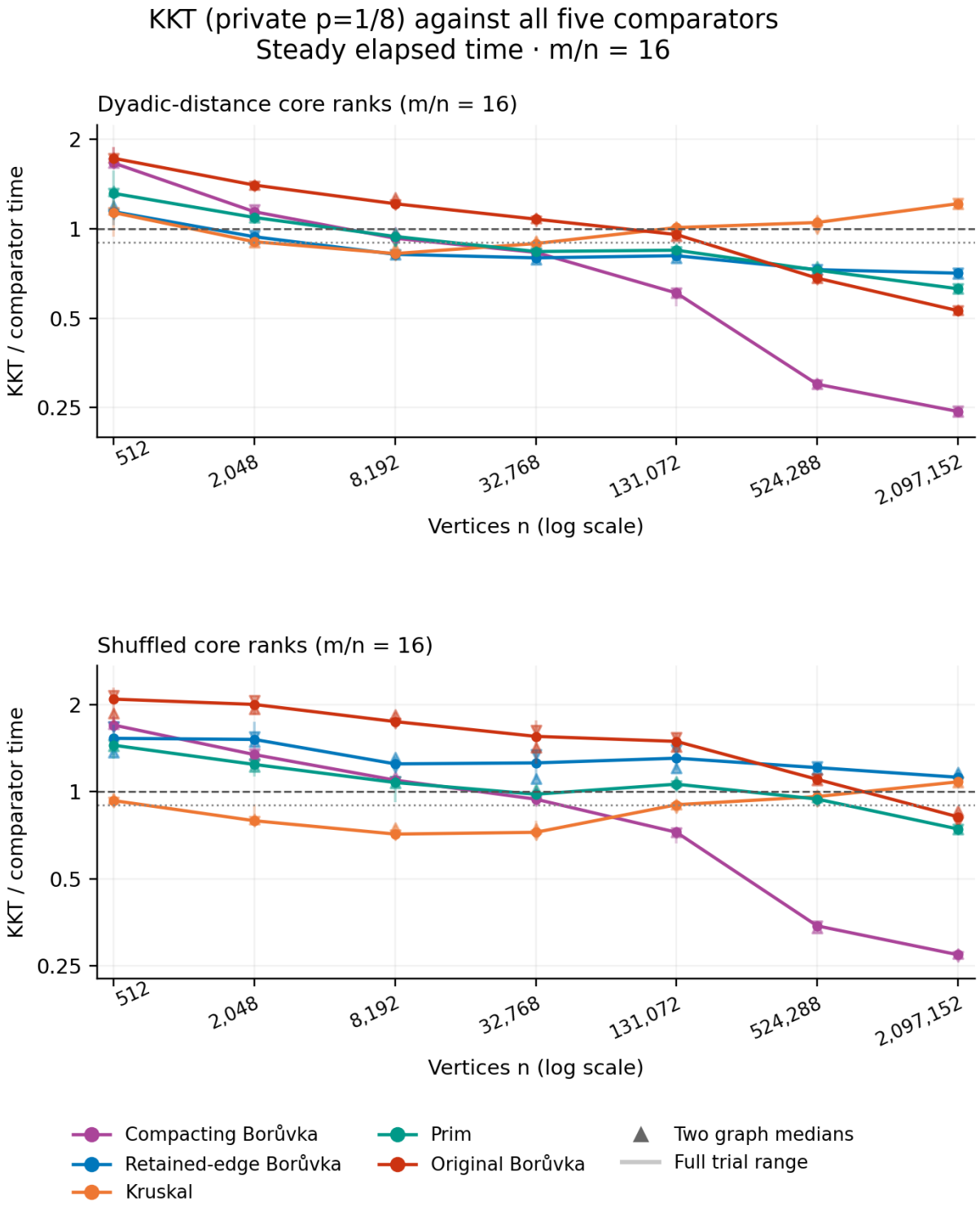
d / weights	Comparator	States in declared size order	Strong observed bracket
16 / distance	Dedup B	L,L,N,S,S,S,S	(2,048, 32,768]
16 / distance	Retained B	L,N,S,S,S,S,S	(512, 8,192]
16 / distance	Kruskal	L,N,S,N,M,L,L	None
16 / distance	Prim	L,L,N,S,S,S,S	(2,048, 32,768]
16 / distance	Original B	L,L,L,L,N,S,S	(32,768, 524,288]
16 / shuffled	Dedup B	L,L,L,N,S,S,S	(8,192, 131,072]
16 / shuffled	Retained B	L,L,L,L,L,L,L	None
16 / shuffled	Kruskal	N,S,S,S,N,N,L	None
16 / shuffled	Prim	L,L,L,M,L,N,S	(131,072, 2,097,152]; final point only
16 / shuffled	Original B	L,L,L,L,L,L,S	(524,288, 2,097,152]; final point only
8 / distance	Dedup B	L,L,L,N,N,S,S,S	(4,096, 262,144]
8 / distance	Retained B	L,L,L,N,N,S,S,M	None
8 / distance	Kruskal	L,L,L,L,L,L,L,L	None
8 / distance	Prim	L,N,S,S,S,S,S,S	(256, 4,096]
8 / distance	Original B	L,L,L,N,S,S,S,S	(4,096, 65,536]

d / weights	Comparator	States in declared size order	Strong observed bracket
8 / shuffled	Dedup B	L,L,L,L,N,S,S,S	(16,384, 262,144]
8 / shuffled	Retained B	L,L,L,L,L,L,L	None
8 / shuffled	Kruskal	L,M,S,S,S,N,L,L	None
8 / shuffled	Prim	L,M,N,S,S,S,S,S	(256, 16,384]
8 / shuffled	Original B	L,L,L,L,L,M,S,S	(65,536, 1,048,576]

Thirteen of the twenty outcomes have a qualifying strong bracket, only one against retained B and none against Kruskal. There are 102 qualifying endpoint pairs among the 490 enumerated endpoint pairs. The two shuffled-d16 brackets ending at 2M have no larger measured point. Earlier KKT leads versus Kruskal reverse; the d8 distance comparison loses to Kruskal at every size. Every current baseline column remains below; each cell is the largest-size pooled **wall/CPU paired-ratio median**, not a ratio of separate runtime medians.

d / weights	Largest n	/ dedup B	/ retained B	/ Kruskal	/ Prim	/ original B
16 / distance	2,097,152	0.2425/0.2425	0.7088/0.7088	1.2152/1.2152	0.6281/0.6281	0.5299/0.5299
16 / shuffled	2,097,152	0.2729/0.2729	1.1219/1.1219	1.0815/1.0815	0.7431/0.7431	0.8186/0.8186
8 / distance	4,194,304	0.3509/0.3509	0.9726/0.9726	1.8485/1.8485	0.5267/0.5267	0.4484/0.4484
8 / shuffled	4,194,304	0.3199/0.3199	1.1558/1.1558	1.3355/1.3355	0.5253/0.5253	0.5419/0.5419

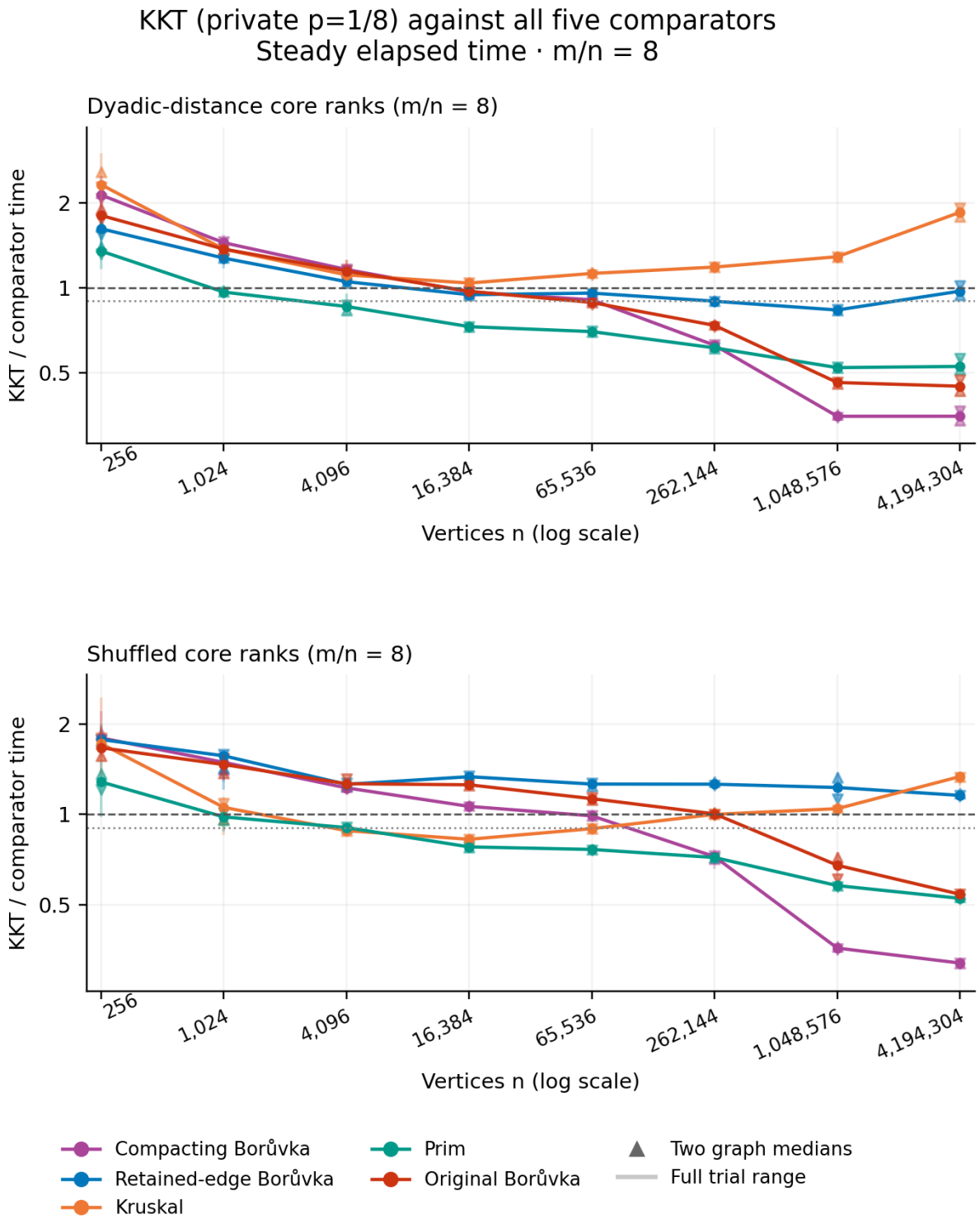
For distance d16, retained-B ratios are **1.142253/1.143671** at 512, **0.940215/0.940332** at 2,048, **0.820461/0.820453** at 8,192 and **0.708792/0.708792** at 2M. The four graph/clock medians move from 1.119160-1.190358 at the losing point to 0.819951-0.821838 at 8,192. All 60 trials across the five strong suffix sizes beat retained B by more than 10% on both clocks. At 2M, all twelve win; wall ratios range 0.685811-0.731127 and CPU 0.685810-0.731122, with graph/clock medians 0.706646-0.711577. This is a clear observed crossover for this implementation and synthetic family, with only two graphs and one repeated sampler.



Density 16: all five comparisons, both weight twins and full trial ranges

For distance d8, the full retained-B sequence is **L,L,L,N,N,S,S,M**. Its pooled ratios improve to 0.896108/0.896118 at 262K and 0.834133/0.834133 at 1M, then deteriorate to **0.972608/0.972607 at 4M**. There only seven of twelve trials win on each clock. Graph/clock medians span 0.941382-1.015940, wall trials 0.931099-1.023289 and CPU 0.931095-1.023291. The pooled value

below one hides a graph-level loss; the final reversal invalidates a sustained strong bracket. Shuffled weights lose all 180 retained-B pairs per clock across the two densities.



Twelve trials per point; two graphs, one sampler. Ranges are descriptive.
Graph two reverses the full schedule; graph and order are confounded.
Lines connect measured sizes only. All 0 flagged calls retained.
Ratios: dashed 1.00 = parity; dotted 0.90 = strong-lead threshold.

Density 8: all five comparisons, both weight twins and full trial ranges

No case is fastest against all five controls by either all graph/clock medians or both pooled clock medians. Seven individual trials per clock do beat all five: five distance-d16 trials at 131,072 and two at 524,288, all on graph 23000029. These isolated wins do not establish a case-level all-six winner. At

every largest endpoint Kruskal wins all twelve pairs. The complete paired table retains all thirty cases, both clocks, graph medians and trial ranges; the point states and bracket table retain every earlier lead and reversal.

All four standalone and eight repaired paper PNGs passed original-resolution review; root independently reviewed the eight paper pages. The first paper attempt had a legend/x-label collision and remains archived. The accepted second attempt changes only presentation margins, preserving exact plotted values. Its two pages above use 7.1×8.8-inch canvases and specified text ≥8.5pt.

Absolute medians give scale. At distance d16/2M, wall times are **2.8306 s KKT, 11.7042 s dedup B, 3.9828 s retained B, 2.3332 s Kruskal, 4.5059 s Prim and 5.3505 s original B**; KKT ranges 2.7586-2.8941 s. At distance d8/4M, they are **4.4784, 12.7942, 4.6048, 2.4387, 8.4535 and 9.9960 s** in the same order, with KKT ranging 4.3265-4.6678 s. Ratios must still come from paired calls. Repetitions reuse inputs; reversed graph order leaves graph/host-history confounding. Historical and current campaigns are not pooled.

What the matched work diagnostics explain

The separate untimed companion repeats all sixty timing fingerprints. Its 180 certified solver outputs passed strict reload and an independent raw audit: all **11,700 counter cells, 120 full comparator chains, 1,236 round rows, 60 query rows and 30 topology twins** agree. Scalar engine words equal sampling-edge visits exactly; unmeasured comparator insertions remain null. The full sixty inputs and all 65 counters are retained, including the timing losses. These diagnostics are not another runtime comparison.

The following largest-size ranges retain both graph seeds. Every row has m=33,554,432. Retained-B input volume is the sum of edge counts over its full contraction chain; KKT input volume sums contraction inputs across its entire recursion. These are named logical counts, not interchangeable units of CPU cost. Q and U are also cumulative over the solve's verifier-tree calls.

d / weights	n	Retained-B rounds	Retained-B input/m	KKT contraction input/m	U/Q
16 / distance	2,097,152	20	17.9059-17.9066	2.4031-2.4037	3.5913-3.5941 %
16 / shuffled	2,097,152	8-9	7.3141-7.9232	2.2889-2.2895	0.9110-0.9203 %
8 / distance	4,194,304	21	17.6875-17.6876	2.5231-2.5233	6.6664-6.7194 %
8 / shuffled	4,194,304	9	7.5173-7.5700	2.2937-2.2943	1.9932-2.0151 %

Ordered weights make retained B revisit a large crossing-edge population through many rounds. Shuffling substantially shortens that chain. KKT itself uses slightly less contraction input on the shuffled twins, yet loses those timing comparisons: which family favors KKT depends on what the competitor avoids as well as what KKT filters. The complete round table and counter table retain every intermediate size, rather than only these endpoints.

The density-8 reversal has an additional measured distinction. At ordered 4M, graph seeds 23000003 and 23000029 have similar KKT contraction-edge visits, about **7.570m**, but verifier-edge visits are **2.7463m and 4.7531m**. Source accounting isolates the wrapper that partitions a disconnected forest into trees: its cumulative visits are **11,478,993 and 78,826,591**. Their difference accounts for the verifier-visit difference up to a −11,760 change in remaining verifier work. The exact partition decomposition covers all sixty inputs.

Further integer bounds identify the root path for these two graphs. The first graph's partition visits are below its 29,884,416 root edges, while total filtered survivors are also below that count; these exclude both a nonempty disconnected root forest and an empty one. Its root forest is therefore connected.

For the second graph, all nonroot calls could contribute at most 20,446,074 partition visits, below the observed 78,826,591, so the root forest must be nonempty and disconnected. Across the full sixty-input matrix, the bounds establish 59 connected roots and this one disconnected root. This is an inference from source identities and aggregate inequalities, not a per-call trace, isolated-vertex count or CPU attribution. Avoiding extra query partitioning and copying on this path is a concrete next implementation target; that change is unimplemented and no gain has been measured.

Source reasoning still gives the ordered root q=n/8 and r=(d−7/8)n after two contractions. With p=1/8, the root light-edge expectation bound q/p is n, or m/d; lower density therefore weakens that normalized bound. The measured cumulative filtered survivors are 0.0878m-0.0881m at the distance-d16 endpoint and 0.1761m-0.1762m at distance-d8. These totals span recursion, not just the root, and the endpoints have different vertex counts. Neither the bound nor this comparison isolates an optimal sampling policy. All six work figures remain supplementary; counters do not measure allocator internals, memory bandwidth or CPU attribution.

12. Correctness, reproducibility and the remaining empirical limits

Independent certificates check source membership, unique IDs, acyclic connectivity and cycle optimality. Every timed/warmup result must match exact unique-MST IDs, plus tuple endpoints/weights where returned. This certifies measured outputs, not undeclared malformed inputs, tie conventions or integer domains.

Separate native/sanitizer gates use independent forest paths and contraction reconstructions; sampling tests cover words, partial blocks, dispatch and public-stream preservation. Wrong-policy/object negatives reject accidental reference dispatch. Final uninstrumented objects are checked separately. Recent timing uses inspected arm64 artifacts on Apple M3 Max/128 GiB, macOS 26.6.2, Apple Clang 15, C++17/O3/native tuning/assertions and 512 MiB stack; Clang 22 ASan/UBSan is a distinct correctness profile. No cross-machine claim follows.

Pre-capture freezes bind cases, seeds, boundaries, source, recipes, objects, commands, binaries and parent proofs. All raw endpoints/results, flags and failed attempts remain. Independent ratio/summary audits avoid collector aggregation; separate identity and numeric mutations test more than hashes. The current Google audit's narrower exclusions are stated in section 4.

Resource monitoring bounds whole owned process groups and requires affirmative empty-group observations after native exit. The larger half-policy capture took **82.4218 minutes** and peaked at **29.7508 GiB sampled group RSS**. Held-out timing took **54.1279 minutes** and sampled at most **8.2377 GiB**. The paired eighth campaign took **65.3455 minutes**, peaked at **8.33675 GiB**, and completed all 48 empty groups with no unavailable RSS observations or clock flags. The completed eighth hierarchy extension took **86.6785 minutes**, peaked at **28.1561 GiB sampled group RSS**, and verified all 28 groups empty with zero unavailable observations or clock flags. The distance campaign took **15.1378 minutes**, sampled **7.4441 GiB** at most across 490 observations, and verified all 24 groups empty with zero unavailable samples or clock flags. The lower-density native campaign took **41.590633 minutes**, peaked at **7.451889 GiB sampled owned-group RSS**, and retained two unavailable observations among 1,341 RSS requests. All sixty native groups and the outer group were verified empty; there were zero clock flags. These are campaign/resource observations, not algorithm runtimes or continuous solver-specific memory peaks. Caffeinate attempts failed; successful power-log queries and absent logged overlap do not prove sleep inhibition, stable thermals or an idle host.

The rejected spatial diagnostic, injected cross-architecture link failure and inaccurate-legend plots remain archived. The spatial repair expands only the frontend allowlist, retaining eight production objects; its fresh matrix reuses no partial output. Corrected plots preserve plotted values.

Small seed counts, order/host confounding and unmeasured hardware effects remain. Counters omit allocator internals, vector relocation, collisions and instructions; uninstrumented final-set insertions are null, not zero. Source, counts and timing retain distinct scopes.

13. What remains open at this cutoff

The old astronomical estimate came from fitted equations with coefficients 4.67 and 2357; their assumed intersection is N=2^(2357/4.67), approximately 9×10^151. That is conditional extrapolation, not an observed threshold or lower bound. Small changes in a coefficient ratio move such an intersection exponentially. Later implementations and stronger comparators cannot inherit those historical constants. Original implementation post.

A strong retained-B advantage now exists at all six measured hierarchy-attachment sizes from 4,096 through 4M and all five measured distance-ranked sizes from 4,096 through 1M; each smallest 1,024-vertex point already narrowly leads. The hierarchy margin weakens at 4M and some individual trials lose. The completed d16 ladder now additionally supplies a losing endpoint and a sustained retained-B bracket; d8 reverses at 4M. What remains open is robustness across more graph/sampler realizations and how broadly the advantage extends beyond these engineered weight arrangements. The six-solver comparison must remain intact: clear wins over retained B do not erase Prim's hierarchy advantage, Kruskal's distance-1M win, or the losses on shuffled and random inputs.

Investigation	Scope	Status at this cutoff
Current Google Benchmark	Three families, two algorithms, eleven sizes, explicit N=n+m	Completed: all 24 exported choices and all 192 custom clock/window choices are N log N; raw arithmetic and figures audited
Lower-density distance/shuffled ladders	Seven density-16 sizes and eight density-8 sizes, all six solvers, fresh seeds	Completed timing/audit: thirteen outcomes have qualifying brackets; distance d16 retained B (512,8,192] is sustained through 2M; distance d8 reverses at 4M; no all-six case winner
Lower-density work companion	Sixty fingerprint-matched inputs, three instrumented solvers and complete comparator chains	Completed: all 11,700 cells and 120 chains independently audited; ordered weights retain more repeated-scan work, and the d8 endpoint exposes differing disconnected-forest overhead
Packed plus compressed	Shared compressed verifier and separate private sampler change	No measured combined gain
Complete hierarchical metric	Complete graphs n=64…16,384, exact metric and matched shuffled-weight twin	Source/math review only; no performance result

The completed seven-size hierarchy ladder, six-size distance ladder and two lower-density ladders are measured evidence above. Future proposals remain separate; a source declaration, readiness receipt or freeze alone is not a measurement. No historical timing is pooled to estimate an unmeasured implementation effect.

The dense metric proposal is also deliberately qualified. Complete topology and compact tier weights preserve dyadic Borůvka rounds while satisfying an ordinary triangle inequality; distinct perturbations make it different from an exact ultrametric. Retained contraction has Θ(m log n) input volume. However, its deduplicated quotient is complete on geometrically fewer vertices, with total volume $(2n^2-3n+1)/3=\Theta(m)$. Even its theory therefore does not predict a six-way win. At $m=\Theta(n^2)$, any complexity axis must use represented input, and a dense-array Prim would be another relevant future implementation.

Unrun proposals do not establish an implementation gain or a new favorable family. The completed evidence already answers the original implementation question: some apparent impracticality came from repairable accounting defects and costly representations. The remaining scientific question concerns which complete algorithm, with which constants and graph reductions, wins against comparably engineered alternatives over an explicitly observed domain.

Evidence and publication notes

This edition cites completed reports and their independently audited canonical tables. Its thirteen embedded figures use accepted paper layouts, while duplicate-clock, absolute-time and historical figures remain explicitly linked below. This is a presentation choice, not removal of numerical evidence. Edition-level rendering, pagination, link verification and page-scale visual checks are recorded separately from the studies' scientific validation.

The packed evidence copy contains the tracked study plus an archived helper, with the copied README's helper link corrected under its receipt. Native build records preserve their original machine paths and object dependencies. Relative publication links and copied figures make the text/evidence collection easier to move; they do not make the historical native proof a self-contained portable rebuild. Native dependency validation and publication link checks have separate scopes.

Complete supplementary figure and table gallery

Every clock and comparator remains available. Images use their original accepted attempts; no timing curve is pooled with a different campaign. The linked study reports retain protocols, raw integers, certificates, audits, full ranges and reproduction commands. SVG/PDF counterparts and original-resolution QA accompany each gallery.

Evidence	Complete figures and both-clock tables
Lower-density ladders	All paired ratios and ranges, all 150 point states, all 490 endpoint pairs, all eight paper figures, complete timing study
Lower-density work	All six figures, all 65 counters, all comparator rounds, query counts, partition decomposition and complete work report
Current complexity	All standalone clocks and models, all six paper layouts, all model fits, complete study
Historical complexity	Normalized figure and historical report; these are not the current opt-in binary
Query compression	Constructed sharing schematic and complete study
Half-policy 4M ladder	Wall ratios, CPU ratios, wall times, CPU times, study
Paired half/eighth	All ten original figures, paper CPU implementation effect, paper CPU direct comparisons, all comparisons
Paired recursive work	Complete audited work report, all 65-field statuses and original six-panel figure; no undefined ratios are dropped
Eighth hierarchy size extension	CPU ratios, wall times, CPU times, all comparisons
Eighth distance ladder	CPU ratios, wall times, CPU times, all comparisons
Earlier family/implementation screens	Common families, retained-edge control, Edge view, density, packed sampler, held-out spatial/random families and held-out work

The current Google native-CPU normalized-time and both model-window pages are included above; its other three clock views and all 1,152 fits remain linked. Lower-density timing and all twelve standalone/paper figures are complete and reviewed. The two wall-ratio pages appear above; both

CPU counterparts and all six-solver absolute-time pages remain in the linked gallery. The lower-density work supplement adds six reviewed figures without expanding the thirteen main figures.

Evidence	Scope and revision anchor
Historical report and common families	Original 693c6a5, repairs and final historical production 2d4fbf1; common infrastructure c9ed4ca
Retention screen	Opt-in source ae03ec6; screen freeze/declaration 36b8ea7
Shared contraction control	Six-solver native/protocol 1bc827d; pre-capture freeze 727d43c
Direct Edge and density	Edge source b7aa02b; separately frozen density reuse
Packed sampling	Original separate-worktree source 4bb476e, freeze/release 610e9c2; publication copy from evidence commit 44f4656, no compressed verifier
Compressed queries	Source a7cb2d6, capture a07f047; only verifier object differs
Held-out families and work follow-up	Timing revision 005d549; separate diagnostic repair/freeze e3997c0 / 1f226729
Larger hierarchy	Frozen unchanged binary SHA-256 74e604830c4c730395cd6c9e9dc61287a6d88a34cc89983bb4ee61dc3272a1b2
Half versus eighth and work follow-up	Opt-in source 116c316; timing capture d45a1c0, evidence bdecf0f; work evidence cb0c208, diagnostic freeze e9b556e7d92b…
Eighth seven-size extension	Exact eighth binary reuse; capture revision cb0c208; freeze SHA-256 aa2f9ae7385105eb7185b864a1f46387f8a02ebe8b967f041d3d0cef5ffa7bc5; timing, audit and standalone visual QA complete
Lower-density timing and work	Timing freeze 0c49ed1d046d402ebccb6978ef5f9cdea02865758a1578a43003b189025aa520, complete 60/2,520 capture and independent arithmetic PASS; separate work freeze ae32f7ffc4cd17c04885d04e6480528882e0b274e61025097206c087a7429ece, complete 60/180 capture and raw-counter audit PASS
Current complexity	Freeze bc0abc5f8675ce9771b6ce81f5cd933afd5d3ffd36d79bc6b34b98e4a89620c7; exact eighth production objects, pinned Google 1.5.2 main/library; capture, independent arithmetic and all eleven figures complete
Eighth distance ladder	Exact eighth binary reuse; capture revision a6bfbdc; freeze SHA-256 3c8866b681c8133dc839dfe55a5e148c93353ee75ce1dc82499aebbed53f338a; timing, audit and original/paper figure QA complete

The held-out and larger campaigns share the half-policy binary identified in the larger row, with separate inputs and declarations; their measurements are not pooled. The larger freeze digest is d4600df2dee18c98d263a9bc043d2881c16a6cc253d9c99047aede4aaa334f27. The paired eighth freeze is 2be7556d15b58f3f65cc380ac22b67c57fdf8aea747d651b9d648c6a1aa59d2f; it uses fresh half/eighth binaries. Each linked report provides reproduction commands, source/build bindings, raw tables, certificates and audits. Reanalysis launches no solver; new native timing remains separately scheduled.